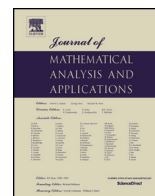




Contents lists available at ScienceDirect

Journal of Mathematical Analysis and Applications

journal homepage: www.elsevier.com/locate/jmaa

Regular Articles

An improved active contour model based on Finsler ∞ -Laplacian [☆]Chenchen Li, Fang Liu ^{*}

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ARTICLE INFO

Article history:

Received 26 July 2025

Available online 24 February 2026

Submitted by G.M. Coclite

Keywords:

Finsler ∞ -Laplacian

ACM

Finsler norm

Image segmentation

ABSTRACT

In this paper, we propose a new active contour model that extends the static external force field under the Finsler norm. The proposed model is formulated as an energy minimization problem, incorporating a data fidelity term and an L^∞ -regularization term based on the Finsler norm. To minimize the energy functional, we consider a second-order singular and degenerate parabolic problem. We establish the comparison principle and demonstrate the existence, Lipschitz continuity in space and uniform continuity in time of the viscosity solution. Experimental results demonstrate that the proposed model achieves superior performance in image segmentation, including a larger capture range, noise robustness and effective preservation of edge details.

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1. Introduction

The field of image processing contains diverse tasks, such as image denoising [2,6,52], image inpainting [9,12] and image segmentation [29,40]. In this paper, we focus on image segmentation, which has numerous applications in object recognition, medical imaging, target tracking and autonomous driving, etc. The main goal is to divide the image into different regions according to the similarity of gray scale, texture and features of pixels. A variety of segmentation methods have been proposed, such as threshold-based methods [42], region-based methods [44], edge detection methods [17], energy functional methods [35–37,40,47], graph theory methods [53] and deep learning methods [15,39].

The active contour model (snake model), one of the classical energy functional methods, was first introduced by Kass [33] et al. in 1988. The model transforms the image segmentation problem into the minimization problem of a curve energy functional. Generally, the active contour models can be classified into parametric active contours [41,48,51] and geometric active contour [28,35–37]. In this paper, we pri-

[☆] This work was supported by National Natural Science Foundation of China (No. 12141104, No. 12526435).

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marily focus on the parametric active contours. Although the introduction of the active contour model was a significant breakthrough, it possesses certain limitations, such as high sensitivity to the initial contour position, a limited capture range of the external force field and poor robustness to noise. To enhance the performance of the activity contour model, many models have been proposed, such as the balloon model [14], etc. Among them, the gradient vector flow (GVF) model introduced by Xu and Prince [50] gained considerable recognition.

By diffusing gradient vectors from the object edges across the entire image, the GVF model effectively expands the capture range and reduces its sensitivity to the initial curve position. Due to its advantages, the GVF model has attracted considerable research attention and spurred the development of many improved variants, such as the GGVF [51], NGVF [41], ILGVF [27], ADF [49] and GVFOM [54] models, etc.

The anisotropic Finsler norm [10] has found extensive applications in image processing. For instance, Esedoğlu and Osher [22] introduced a Finsler norm to replace the regularization term $|Du|$ in the classical total variation (TV) denoising model, enabling the model to preferentially preserve specific edge orientations while performing denoising. Balana and Stojanov [5] proposed a Beltrami flow based on the Finsler norm for evolving image surfaces, which demonstrates sensitivity to directional structures, such as edges. Compared to the isotropic cases, their experiments also demonstrated that the anisotropic Finsler norms selectively process noise based on pixel directionality, thereby reducing noise impact while preserving edges and important features.

In this paper, inspired by the aforementioned studies, we propose a new active contour model by introducing the anisotropic Finsler norm. The anisotropy of the Finsler norm enables our model to drive the contour evolution based on local directional information in the image, thereby effectively preserving edge details while exhibiting excellent noise robustness.

We formulate the Euler-Lagrange equation under the Finsler norm and study the solutions in the viscosity sense in Section 3. We initially establish the comparison principle of the corresponding parabolic equation by the method of double variables, overcoming the challenge of non-differentiability at the origin. Subsequently, by the Perron method, we prove the existence of the viscosity solution. Furthermore, we demonstrate the Lipschitz continuity of the viscosity solution in space using the method of the continuous modulus and show the uniform continuity of the viscosity solution in time by the mollifier argument. Although our model is two-dimensional, these results still hold for $n \geq 2$. To discretize the parabolic equation, we adopt a central difference scheme for the diffusion term and an upwind scheme for the convection term. By the criteria of Barles and Souganidis in [8], it is easy to prove that the approximate solution is convergent to a function u . The segmentation results demonstrate that our model offers distinct advantages, such as a larger capture range, insensitivity to the initial contour, robustness to noise and superior edge detail preservation.

The organization of this paper is as follows. Section 2 reviews the necessary background knowledge, including the classical snake model and several types of GVF models. In Section 3, we provide a detailed description of the proposed model, along with the definitions of the viscosity solution and statements of the main theorems. We establish the comparison principle of Problem (3.2) using the method of double variables and then prove the existence of the viscosity solution u by the Perron method. Moreover, we demonstrate that u is Lipschitz continuous in space and uniformly continuous in time. Section 4 presents numerical experiments validating the proposed model. Additionally, comparative tests with other models confirm its improved robustness to noise and enhanced preservation of edge details.

2. Backgrounds

This section provides an overview of the active contour model and discusses several improved variants, including the GVF, GGVF, NGVF, ILGVF, ADF and GVFOM models.

2.1. Active contour model

In the seminal work by Kass [33] et al., the active contour model (Snake) is defined as a parametric curve $c(s) := [x(s), y(s)]$ ($s \in [0, 1]$), which minimizes an energy functional

$$I[c(s)] := \int_0^1 \left\{ \frac{1}{2} (\alpha |c'(s)|^2 + \beta |c''(s)|^2) + E_{\text{ext}}(c(s)) \right\} ds.$$

Here, $c'(s)$ and $c''(s)$ are the first and second derivative of $c(s)$, $|\cdot|$ denotes the Euclidean norm and α, β are the weighted parameters. The first term denotes the internal force of the image, which controls the continuous evolution of the curve $c(s)$ and ensures its smoothness. The second term $E_{\text{ext}}(c(s))$ represents the external force of the image, which drives $c(s)$ to the object edges and exhibits lower values at the edges. Typical $E_{\text{ext}}(c(s))$ for a grayscale image $I(x, y)$ are defined as follows

$$E_{\text{ext}}(x, y) := -|\nabla I(x, y)|^2,$$

and

$$E_{\text{ext}}(x, y) := -|\nabla[G_\sigma(x, y) * I(x, y)]|^2.$$

2.2. GVF: gradient vector flow

Based on the active contour model, Xu and Prince proposed the gradient vector flow model (GVF) [50], which minimizes the following energy functional

$$I[V] := \iint_{\Omega} \mu (u_x^2 + u_y^2 + v_x^2 + v_y^2) + |Df|^2 |V - Df|^2 dx dy,$$

where Ω is a bounded region of $\mathbb{R}^2$, μ represents the weighted parameter, Df is the gradient of the edge image f and $V(x, y) := [u(x, y), v(x, y)]$ denotes the GVF vector field. The GVF model generates a dynamic force field through the diffusion along the edges, which allows the gradient vectors to extend into more distant regions. By the variational method, they obtained the following Euler-Lagrange equations

$$\begin{cases} \mu \Delta u(x, y) - 2|Df|^2(u(x, y) - f_x) = 0 & (x, y) \in \Omega, \\ \mu \Delta v(x, y) - 2|Df|^2(v(x, y) - f_y) = 0 & (x, y) \in \Omega, \end{cases}$$

where Δ is the Laplacian. The evolution of the Euler-Lagrange equations is achieved by treating $V(x, y)$ as a function of time t and embedding it into a dynamic scheme. Compared to the Snake model, the GVF model exhibits a significantly larger capture range. Moreover, it acquires a notable capability to converge into concave regions.

2.3. GGVF: generalized gradient vector flow

Although the GVF model improves the limitations of the Snake model, it still faces challenges in preserving weak boundaries and converging into concave regions. In [51], Xu and Prince further extended the GVF model to propose the generalized gradient vector flow (GGVF) model and considered the following system

$$\begin{cases} u_t(x, y) := k(|Df|)\Delta u(x, y) - h(|Df|)(u(x, y) - Df) & (x, y) \in \Omega, \\ v_t(x, y) := k(|Df|)\Delta v(x, y) - h(|Df|)(v(x, y) - Df) & (x, y) \in \Omega, \end{cases}$$

where $k(|Df|) := e^{-(|Df|/K)}$ (K is a threshold parameter) and $h(|Df|) := 1 - k(|Df|)$. The GGVF model can converge to narrow and deep concavities and demonstrate better robustness in noisy images.

2.4. NGVF: normal gradient vector flow

Ning et al. [41] proposed the normal gradient vector flow (NGVF) model, defined by the following equations

$$\begin{cases} u_t(x, y) := \mu \Delta_{\infty}^N u(x, y) - |Df|^2(u(x, y) - f_x) & (x, y) \in \Omega, \\ v_t(x, y) := \mu \Delta_{\infty}^N v(x, y) - |Df|^2(v(x, y) - f_y) & (x, y) \in \Omega, \end{cases}$$

where $\Delta_{\infty}^N u := |Du|^{-2}(u_x^2 u_{xx} + 2u_x u_y u_{xy} + u_y^2 u_{yy})$ is called the normalized ∞ -Laplacian. Comparing with the GVF model, the NGVF model has a better ability to access narrow and deep concavities. In certain cases, a larger time step can make the NGVF model more effective than the GVF model.

2.5. ILGVF: infinity-Laplacian gradient vector flow

Carole et al. [27] proposed the infinity-Laplacian gradient vector flow (ILGVF) model, which minimizes the energy functional

$$I[v] := \iint_{\Omega} |Dv - U|^2 |U|^2 dx dy + \mu \|Dv\|_{L^{\infty}(\Omega)}, \quad v \in W^{1,\infty}(\Omega),$$

where U denotes the gradient vector field and $\|\cdot\|_{L^{\infty}}$ is the L^{∞} -norm. The ILGVF model reduces to a single unknown v by modeling the vector field as a scalar potential gradient, extending capture range.

2.6. ADF: adaptive diffusion flow

In [49], Wu et al. proposed the adaptive diffusion flow (ADF) model, which minimizes the energy functional

$$\begin{aligned} I[V] := & \iint_{\Omega} k(|Df|) \left(-m \|\Theta\|_{L^{\infty}(\Omega)} + (1-m) \frac{1}{p(|Df|)(\sqrt{1+\Theta})^{p(|Df|)}} \right) dx dy \\ & + \iint_{\Omega} h(|Df|) (|V - Df|^2) dx dy, \end{aligned}$$

where m, k, h, p are the weighted functions and $\Theta := |G_{\sigma} \otimes DV|^2$ (G_{σ} is the Gaussian kernel of standard deviation σ). The ADF model outperforms the GVF model in convergence, as ∞ -Laplacian efficiently guides the contour into long, narrow concavities within homogeneous image regions.

2.7. GVFORM: gradient vector flow over manifold

Zhang et al. [54] proposed the gradient vector flow over manifold (GVFORM) model, in which the Laplacian of the GVF model is replaced by the operator Δ_g and the corresponding Euler-Lagrange equations are given by

$$\begin{cases} \mu \Delta_g u(x, y) - (u(x, y) - f_x)(f_x^2 + f_y^2) = 0 & (x, y) \in \Omega, \\ \mu \Delta_g v(x, y) - (v(x, y) - f_y)(f_x^2 + f_y^2) = 0 & (x, y) \in \Omega, \end{cases}$$

where $\Delta_g u := \frac{1}{\sqrt{g}} \partial_u (\sqrt{g} g^{ij} \partial_v u)$ ($i, j = 1, 2$), $\Delta_g v := \frac{1}{\sqrt{g}} \partial_u (\sqrt{g} g^{ij} \partial_v v)$, g represents the determinant of the metric matrix (g_{ij}) and (g^{ij}) denotes the inverse matrix of (g_{ij}) . The GVFO model shows efficiency in narrow and deep concavity regions, weak edges and noise robustness.

3. Proposed method

Inspired by former works, we introduce a new model based on the Finsler norm by minimizing the energy functional

$$I[u] := \int_{\Omega} H(Du - w)^2 H(w)^2 dx dy + \mu \|H(Du)\|_{L^\infty(\Omega)}, \quad u \in W^{1,\infty}(\Omega), \quad (3.1)$$

where, $\mu > 0$ is a weighted parameter, Ω denotes a bounded region of $\mathbb{R}^2$, $\|\cdot\|_{L^\infty}$ is the L^∞ -norm, f represents the edge image and $w = (w_1, w_2) := -Dg(|Df|)$ is the gradient vector flow. The edge detector function $g : [0, +\infty) \rightarrow [0, +\infty)$ is monotonically decreasing, satisfies $g(0) = 1$ and $\lim_{s \rightarrow +\infty} g(s) = 0$. For example, $g(s) = \frac{1}{1+s^2}$. In homogeneous regions, where $|Df| \rightarrow 0^+$, we conclude that $g(|Df|) = 1$ and $w = 0$. At edges, as $|Df| \rightarrow +\infty$, the detector function satisfies $g(|Df|) = 0$. Here and after, $H : \mathbb{R}^2 \rightarrow [0, +\infty)$ is a Finsler norm satisfying the following conditions

$H_1(\text{Regularity})$: $H \in C^2(\mathbb{R}^2 \setminus \{0\})$.

$H_2(\text{Absolute homogeneity})$: H is absolutely homogeneous of degree 1, i.e.

$$H(t\xi) = |t|H(\xi), \quad \forall \xi \in \mathbb{R}^2, \forall t \in \mathbb{R}.$$

$H_3(\text{Strong Convexity})$: $D_\xi^2(H^2)(\xi) > 0_2$ on $\mathbb{R}^2 \setminus \{0\}$.

The main work of this paper is to introduce the anisotropic Finsler norm to the active contour model, which is closely related to the Finsler ∞ -Laplacian equation. The anisotropic Finsler ∞ -Laplacian is strongly degenerate, which makes the uniformly elliptic theories do not work. Moreover, the introduction of the Finsler norm makes the structure of Finsler infinity-Laplacian be more complex. We establish the comparison principle for Problem (3.2) by the method of double variables under the Finsler norm, overcoming the challenge of non-differentiability at the origin. By the Perron method, we prove the existence of a viscosity solution u to Problem (3.2). Moreover, we present the Lipschitz continuity of u in space using the method of the continuous modulus and show the uniform continuity of u in time by the mollifier argument. By the criteria of Barles and Souganidis in [8], it is easy to prove that the approximate solution is convergent to a function u . Experimental results demonstrate that the proposed model achieves superior performance in image segmentation, including a larger capture range, noise robustness and effective preservation of edge details.

Remark 3.1. The functional (3.1) is defined on a bounded domain $\Omega \in \mathbb{R}^2$, in this case, boundary conditions must be specified. Neumann boundary conditions are appropriate for image processing applications, as it effectively reflects the data along the edges of the region. Consequently, it is no longer necessary to define boundary values.

Remark 3.2. If u is a minimizer of the functional (3.1) in Ω , then $u + C$ (C is any real constant) is also a minimizer. This is not an issue, as our primary interest lies in the gradient vector field Du , rather than

the specific value of u . Since $u \in W^{1,\infty}(\Omega)$, it follows that u is Lipschitz continuous. By the Rademacher's theorem [23], u is differentiable almost everywhere in Ω .

Remark 3.3. Since $L^\infty(\Omega) \subset L^2(\Omega)$ holds for any bounded region Ω , it follows that $Du \in L^2(\Omega, \mathbb{R}^2)$. Due to the definition of the gradient vector flow $w = (w_1, w_2) = -Dg(|Df|)$ and the fact that the function g is bounded, we conclude $w \in L^2(\Omega, \mathbb{R}^2)$. Therefore, the integrability of the terms in the first part of the functional (3.1) is ensured.

In order to minimize the energy functional (3.1), we first derive the Euler-Lagrange equation by the variational method. The Euler-Lagrange equation is given by

$$2\Delta_H(u - W)H(w)^2 + 2\langle D_\xi(H(w)^2), H(Du - w)D_\xi H(Du - w) \rangle + \mu\Delta_{H;\infty}^N u = 0,$$

where $\Delta_H(u - W) := \operatorname{div}(H(Du - w)D_\xi H(Du - w))$ ($DW := w$) is called the Finsler Laplacian and $\Delta_{H;\infty}^N u := \sum_{i,j=1}^2 \frac{\partial H}{\partial \xi_i}(Du) \frac{\partial^2 u}{\partial x_i \partial x_j} \frac{\partial H}{\partial \xi_j}(Du)$ is called the Finsler ∞ -Laplacian.

For the case $H(\xi) = |\xi|$, the operator $\Delta_{H;\infty}^N u$ reduces to the normalized ∞ -Laplacian $\Delta_\infty^N u$. The normalized ∞ -Laplacian was introduced by Peres, Schramm, Sheffield and Wilson [43] while studying a class of two person zero-sum games known as Tug-of-War. They proved that the continuous value function of the game is the unique viscosity solution to the following normalized ∞ -Laplacian problem

$$\begin{cases} \Delta_\infty^N u = f_1 & x \in \Omega, \\ u = g & x \in \partial\Omega, \end{cases}$$

where f_1 is the running payment function and g is the final payment function.

The ∞ -Laplacian $\Delta_\infty u := \sum_{i,j=1}^2 \frac{\partial u}{\partial x_i} \frac{\partial^2 u}{\partial x_i \partial x_j} \frac{\partial u}{\partial x_j}$ was first introduced by Aronsson [3] in the 1960's while studying absolute minimizing Lipschitz extension (AMLE) problem. Owing to the high degeneracy and non-divergence form of the operator $\Delta_\infty u$, the solutions are generally understood within the viscosity framework introduced by Crandall, Ishii and Lions [16]. The viscosity solutions of the equation $\Delta_\infty u = 0$ are referred to as ∞ -harmonic functions [31].

The ∞ -Laplacian has been widely applied in image processing [6,11,13,18]. Furthermore, it has been extended to weighted graphs by appropriate discretizations approximating continuous partial differential operators, yielding the graph ∞ -Laplacian [1,19–21]. Subsequently, this development has significantly broadened the application scope of the non-local ∞ -Laplacian, extending its reach from image processing to machine learning [25], graph neural networks [26,46] and other related fields.

This paper focuses on image segmentation. A function $u \in W^{1,\infty}$ is initially defined on the bounded region $\Omega = [0, 1] \times [0, 1]$. We first extend u to $[-1, 1] \times [-1, 1]$ by symmetry and then periodically extend u to the entire plane $\mathbb{R}^2$ [4]. Therefore, for any $h \in \mathbb{Z}^2$ and $x \in \mathbb{R}^2$, the condition $u(x + 2h) = u(x)$ holds. To model dynamic evolution, we introduce the time variable t and transform the Euler-Lagrange equation to a time-evolving parabolic problem

$$\begin{cases} \frac{\partial u}{\partial t} = m(x)\Delta_H(u - W) - \langle q(x), H(Du - w)D_\xi H(Du - w) \rangle + \mu\Delta_{H;\infty}^N u & \text{in } \mathbb{R}^2 \times (0, T), \\ u(x, 0) = u_0(x) & \text{in } \mathbb{R}^2, \end{cases} \quad (3.2)$$

where $C(\mathbb{R}^2) \ni m(x) := 2H(w(x))^2$, $C(\mathbb{R}^2) \cap W^{1,\infty}(\mathbb{R}^2) \ni q(x) := -2D_\xi(H(w(x)))^2$ and $u_0 \in C(\mathbb{R}^2) \cap W^{1,\infty}(\mathbb{R}^2)$ denotes the initial condition. The function $m(x)$ satisfies $\|m(x)\|_{L^\infty(\mathbb{R}^2)} = \zeta_m$ and $m(x)^{\frac{1}{2}}$ is Lipschitz continuous with the Lipschitz constant k_m . Similarly, $q(x)$ has a Lipschitz constant k_q and satisfies $\|q(x)\|_{L^\infty(\mathbb{R}^2)} = \zeta_q$.

3.1. Definitions of viscosity solutions

Now we provide the definitions of viscosity solutions for Problem (3.2). Due to the singularity of the operator $\Delta_{H;\infty}^N u$ at points where the gradient vanishes, we adopt the viscosity solution framework [8,16,30,38].

The equation in Problem (3.2) can be rewritten as

$$\frac{\partial u}{\partial t} + F(x, Du, D^2u) = 0, \quad (3.3)$$

where $F : \mathbb{R}^2 \times \mathbb{R}^2 \setminus \{0\} \times S^2$ (S^2 denotes the collection of 2×2 symmetric matrices with real entries) is defined by

$$F(x, p, X) := b(x, p) + R(x, p, X) + Q(p, X),$$

and the components are defined as follows

$$\begin{aligned} b(x, p) &:= \langle q(x), H(p - w) D_\xi H(p - w) \rangle, \\ R(x, p, X) &:= -m(x) \operatorname{div}(H(p - w) D_\xi H(p - w)) + H(p - w) D_\xi^2 H(p - w)(X - Dw), \end{aligned}$$

and

$$Q(p, X) := -\mu \langle X D_\xi H(p), D_\xi H(p) \rangle.$$

We define the set of upper semicontinuous functions as

$$\operatorname{USC}(\mathbb{R}^2 \times [0, T)) := \{u : \mathbb{R}^2 \times [0, T) \rightarrow \mathbb{R} \mid \text{locally bounded, upper semicontinuous}\},$$

and the set of lower semicontinuous functions as

$$\operatorname{LSC}(\mathbb{R}^2 \times [0, T)) := \{u : \mathbb{R}^2 \times [0, T) \rightarrow \mathbb{R} \mid \text{locally bounded, lower semicontinuous}\}.$$

It is easy to verify the following properties

P₁ : The operator F is elliptic, i.e.

$$F(x, p, X) \geq F(x, p, Y) \text{ for any } p \in \mathbb{R}^2 \setminus \{0\}, X, Y \in S^2 \text{ and } X \leq Y.$$

P₂ : The function Q is continuous on $\mathbb{R}^2 \setminus \{0\} \times S^2$ and locally bounded on $\mathbb{R}^2 \times S^2$. Additionally,

$$Q^*(0, 0) = Q_*(0, 0) = 0,$$

where Q^* (resp. Q_*) denotes the USC (resp. LSC)-envelope of Q .

P₃ : For all $x, y \in \mathbb{R}^2$, the function $b(x, p)$ is locally Lipschitz continuous and

$$|b(x, p) - b(y, p)| \leq k_q |H(p - w) D_\xi H(p - w)| H^*(x - y),$$

where k_q is the Lipschitz constant of the function $q(x)$ and $H^*(x)$ is the dual norm of the Finsler norm H [38].

Although our model in this paper is presented for the case $n = 2$, in fact, the results are valid for any dimension $n \geq 2$. If a function $\phi(x, t)$ is twice differentiable with respect to x in $\mathbb{R}^n \times [0, T)$, we define

$$\Delta_{H;\infty}^+ \phi(x, t) := \begin{cases} \langle D^2 \phi(x, t) D_\xi H(D\phi(x, t)), D_\xi H(D\phi(x, t)) \rangle & \text{if } D\phi(x, t) \neq 0, \\ \max \{ \langle D^2 \phi(x, t) e, e \rangle : H^*(e) = 1 \} & \text{if } D\phi(x, t) = 0, \end{cases}$$

and

$$\Delta_{H;\infty}^- \phi(x, t) := \begin{cases} \langle D^2 \phi(x, t) D_\xi H(D\phi(x, t)), D_\xi H(D\phi(x, t)) \rangle & \text{if } D\phi(x, t) \neq 0, \\ \min \{ \langle D^2 \phi(x, t) e, e \rangle : H^*(e) = 1 \} & \text{if } D\phi(x, t) = 0. \end{cases}$$

Thus, the USC-envelope and LSC-envelope of F are given by

$$F^+(x, D\phi(x, t), D^2\phi(x, t)) := b(x, D\phi(x, t)) + R(x, D\phi(x, t), D^2\phi(x, t)) + Q^+(D\phi(x, t), D^2\phi(x, t)),$$

and

$$F^-(x, D\phi(x, t), D^2\phi(x, t)) := b(x, D\phi(x, t)) + R(x, D\phi(x, t), D^2\phi(x, t)) + Q^-(D\phi(x, t), D^2\phi(x, t)),$$

respectively, where $Q^+(D\phi(x, t), D^2\phi(x, t)) := -\mu \Delta_{H;\infty}^+ \phi(x, t)$ and $Q^-(D\phi(x, t), D^2\phi(x, t)) := -\mu \Delta_{H;\infty}^- \phi(x, t)$.

Let $u \prec_{(x_0, t_0)} \phi$ denote that the function $u - \phi$ attains a maximum at (x_0, t_0) . Similarly, let $u \succ_{(x_0, t_0)} \phi$ represent that $u - \phi$ attains a minimum at (x_0, t_0) .

Definition 3.4. A function $u \in \text{USC}(\mathbb{R}^n \times [0, T))$ is called a viscosity subsolution of the equation (3.3) in $\mathbb{R}^n \times [0, T)$, if for any $(x_0, t_0) \in \mathbb{R}^n \times [0, T)$ and $\phi \in C^{2,1}(\mathbb{R}^n \times [0, T))$ such that $u \prec_{(x_0, t_0)} \phi$, there holds

$$\phi_t(x_0, t_0) + F^+(x_0, D\phi(x_0, t_0), D^2\phi(x_0, t_0)) \leq 0.$$

Similarly, a function $v \in \text{LSC}(\mathbb{R}^n \times [0, T))$ is called a viscosity supersolution of the equation (3.3) in $\mathbb{R}^n \times [0, T)$, if for any $(x_0, t_0) \in \mathbb{R}^n \times [0, T)$ and $\phi \in C^{2,1}(\mathbb{R}^n \times [0, T))$ such that $u \succ_{(x_0, t_0)} \phi$, there holds

$$\phi_t(x_0, t_0) + F^-(x_0, D\phi(x_0, t_0), D^2\phi(x_0, t_0)) \geq 0.$$

A function u is called a viscosity solution of equation (3.3) in $\mathbb{R}^n \times [0, T)$, if u is both a viscosity subsolution and a viscosity supersolution of the equation (3.3) in $\mathbb{R}^n \times [0, T)$.

Viscosity solutions can be defined equivalently using the super-jet and sub-jet [16,34].

Definition 3.5. Let $u : \mathbb{R}^n \times [0, T) \rightarrow \mathbb{R}$ and $(x_0, t_0) \in \mathbb{R}^n \times [0, T)$. If there exists $(a, p, X) \in \mathbb{R} \times \mathbb{R}^n \times S^n$ such that

$$u(x, t) \leq u(x_0, t_0) + a(t - t_0) + \langle p, x - x_0 \rangle + \frac{1}{2} \langle X(x - x_0), x - x_0 \rangle + o((t - t_0) + H^*(x - x_0)^2),$$

where $(x, t) \in \mathbb{R}^n \times [0, T) \rightarrow (x_0, t_0)$, then we write $(a, p, X) \in \mathcal{P}^{2,+}u(x_0, t_0)$ and the set $\mathcal{P}^{2,+}u(x_0, t_0)$ is called the second-order parabolic super-jet of u at the point (x_0, t_0) . Its closure is defined by

$$\begin{aligned} \overline{\mathcal{P}}^{2,+}u(x_0, t_0) := & \left\{ (a, p, X) \in \mathbb{R} \times \mathbb{R}^n \times S^n : \exists (x_n, t_n, a_n, p_n, X_n) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^n \right. \\ & \times S^n, \text{ s.t. } (a_n, p_n, X_n) \in \mathcal{P}^{2,+}u(x_n, t_n) \text{ and } (x_n, t_n, u(x_n, t_n), a_n, p_n, X_n) \\ & \left. \rightarrow (x_0, t_0, u(x_0, t_0), a, p, X) \right\}. \end{aligned}$$

Similarly, we define the second-order parabolic sub-jet of $u(x_0, t_0)$ is $\mathcal{P}^{2,-}u(x_0, t_0) := -\mathcal{P}^{2,+}(-u)(x_0, t_0)$ and its closure is given by

$$\begin{aligned} \overline{\mathcal{P}}^{2,-}u(x_0, t_0) := & \left\{ (a, p, X) \in \mathbb{R} \times \mathbb{R}^n \times S^n : \exists (x_n, t_n, a_n, p_n, X_n) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}^n \right. \\ & \times S^n, \text{ s.t. } (a_n, p_n, X_n) \in \mathcal{P}^{2,-}u(x_n, t_n) \text{ and } (x_n, t_n, u(x_n, t_n), a_n, p_n, X_n) \\ & \left. \rightarrow (x_0, t_0, u(x_0, t_0), a, p, X) \right\}. \end{aligned}$$

Definition 3.6. A function $u \in \text{USC}(\mathbb{R}^n \times [0, T])$ is called a viscosity subsolution of the equation (3.3) in $\mathbb{R}^n \times [0, T]$, if

$$a + F^+(x_0, p, X) \leq 0 \quad \text{for any } (x_0, t_0) \in \mathbb{R}^n \times [0, T] \text{ and } (a, p, X) \in \overline{\mathcal{P}}^{2,+}u(x_0, t_0).$$

Similarly, a function $u \in \text{LSC}(\mathbb{R}^n \times [0, T])$ is called a viscosity supersolution of the equation (3.3) in $\mathbb{R}^n \times [0, T]$, if

$$a + F^-(x_0, p, X) \geq 0 \quad \text{for any } (x_0, t_0) \in \mathbb{R}^n \times [0, T] \text{ and } (a, p, X) \in \overline{\mathcal{P}}^{2,-}u(x_0, t_0).$$

A function u is called a viscosity solution of (3.3) in $\mathbb{R}^n \times [0, T]$, if u is both a viscosity subsolution and a viscosity supersolution of (3.3) in $\mathbb{R}^n \times [0, T]$.

3.2. Comparison principle, existence and regularity

We establish the comparison principle of Problem (3.2) by the method of double variables in $\mathbb{R}^n \times [0, T]$. Subsequently, we employ Perron method to demonstrate the existence of the viscosity solution u . Then, the Lipschitz continuity of u in space is proven by the method of the continuous modulus. We also establish the uniformly continuity of u in time using the mollifier argument. We extend the idea in [27] where they considered the isotropic ∞ -Laplacian. However, the construction of barrier functions is more complex and involves more intricate calculations.

Lemma 3.7. Let $w(x) \in L^2(\mathbb{R}^n, \mathbb{R}^n)$. For each $t \in [0, T]$, if $u(\cdot, t) \in W^{1,\infty}(\mathbb{R}^n)$ and $\|q(x)\|_{L^\infty} = \zeta_q$ ($x \in \mathbb{R}^n$), then there exist positive constants $C_0(w, H, t)$ and $C_1(w, H, D_\xi H, t)$ such that

$$\langle q(x), H(Du(\cdot, t) - w)D_\xi H(Du(\cdot, t) - w) \rangle \leq \zeta_q C_0 C_1.$$

Proof. By [38], there exist constants $0 < \rho \leq \sigma$ satisfying

$$\rho|\xi| \leq H(\xi) \leq \sigma|\xi| \text{ for all } \xi \in \mathbb{R}^n.$$

By Remark 3.3, it follows that $w \in L^\infty(\mathbb{R}^n, \mathbb{R}^n)$. Since $u(\cdot, t) \in W^{1,\infty}(\mathbb{R}^n)$, the gradient $Du(\cdot, t) \in L^\infty(\mathbb{R}^n, \mathbb{R}^n)$. By Property H_1 , we have

$$H(Du(\cdot, t) - w) \leq C_0 \text{ and } |D_\xi H(Du(\cdot, t) - w)| \leq C_1,$$

where C_0, C_1 are positive constants. Recalling Lemma 3.2 in [38], one has

$$\langle q(x), H(Du(\cdot, t) - w) D_\xi H(Du(\cdot, t) - w) \rangle \leq \zeta_q C_0 C_1. \quad \square$$

Theorem 3.8 (comparison principle). *Let $u \in \text{USC}(\mathbb{R}^n \times [0, T])$ be a bounded and periodic viscosity subsolution of Problem (3.2), with the same periodicity as the initial condition $u_0 \in C(\mathbb{R}^n) \cap W^{1,\infty}(\mathbb{R}^n)$ and $v \in \text{LSC}(\mathbb{R}^n \times [0, T])$ be a bounded and periodic viscosity supersolution of Problem (3.2), with the same periodicity as the initial condition $v_0 \in C(\mathbb{R}^n) \cap W^{1,\infty}(\mathbb{R}^n)$ in $\mathbb{R}^n \times [0, T]$. If $u_0(x) \leq v_0(x)$ for all $x \in \mathbb{R}^n$, then $u \leq v$ in $\mathbb{R}^n \times [0, T]$.*

For the reader's convenience, the main steps of proof can be found in Appendix.

With the comparison principle in hand, we can give the following existence result by the Perron method. For the parabolic ∞ -Laplacian, one can see [27, 32].

Theorem 3.9. *Let $u_0 \in C(\mathbb{R}^n) \cap W^{1,\infty}(\mathbb{R}^n)$ be a bounded and periodic initial solution of Problem (3.2) in $\mathbb{R}^n \times [0, T]$, there exists a bounded and periodic viscosity solution of Problem (3.2) in $\mathbb{R}^n \times [0, T]$, which keeps the same periodicity as u_0 .*

The proof of Theorem 3.9 is provided in the Appendix.

To establish the Lipschitz continuity of u in space on $\mathbb{R}^n \times [0, T]$, we employ the method of the continuous modulus. For the isotropic case $F(\xi) = |\xi|$, one can refer to [27].

Theorem 3.10 (Lipschitz continuity in space). *Let $u(x, t)$ be a viscosity solution of Problem (3.2) in $\mathbb{R}^n \times [0, T]$ with bounded and periodic initial condition $u_0 \in C(\mathbb{R}^n) \cap W^{1,\infty}(\mathbb{R}^n)$. If*

$$\|Du_0\|_{L^\infty(\mathbb{R}^n)} \leq E_0$$

for a positive constant E_0 , then there holds

$$\|Du(\cdot, t)\|_{L^\infty(\mathbb{R}^n)} \leq E(t) = E_0 e^{\beta t} + C \frac{e^{\beta t} - 1}{\beta}, \quad t \in [0, T], \quad (3.4)$$

where $C = C(w, H, D_\xi H, k_q)$ and $\beta = \beta(w, D_\xi H, D_\xi^2 H, k_m)$ are constants.

For the proof of Theorem 3.10, see the Appendix.

To prove the uniform continuity of $u(x, t)$ in time on $\mathbb{R}^n \times [0, T]$, we employ a mollifier argument. We first recall the definition and properties of the mollifier [4, 24].

Definition 3.11. (i) Let $\rho \in C_c^\infty(\mathbb{R}^n)$ be defined by

$$\rho(x) := \begin{cases} C \exp\left(\frac{1}{|x|^2 - 1}\right) & \text{if } |x| < 1, \\ 0 & \text{if } |x| \geq 1, \end{cases}$$

where the constant $C > 0$ satisfies

$$\int_{\mathbb{R}^n} \rho(x) dx = 1.$$

The function ρ is called the standard mollifier.

(ii) For each $\varepsilon > 0$, the mollified functions are defined as

$$\rho_\varepsilon(x) := \frac{1}{\varepsilon^n} \rho\left(\frac{x}{\varepsilon}\right).$$

These functions belong to $C_c^\infty(\mathbb{R}^n)$ and satisfy

$$\int_{\mathbb{R}^n} \rho_\varepsilon(x) dx = 1, \quad \text{spt}(\rho_\varepsilon) \subset B(0, \varepsilon),$$

where $B(0, \varepsilon)$ denotes the ball of radius ε centered at the origin and $\text{spt}(\rho_\varepsilon)$ represents the support of ρ_ε .

Definition 3.12. Let $u : \Omega \rightarrow \mathbb{R}$ be a locally integrable function and $\Omega_\varepsilon := \{x \in \Omega \mid \text{dist}(x, \partial\Omega) > \varepsilon\}$. The mollification of u is defined as

$$u^\varepsilon(x) := \rho_\varepsilon * u(x) = \int_{\Omega} \rho_\varepsilon(x - y) u(y) dy = \int_{B(0, \varepsilon)} \rho_\varepsilon(y) u(x - y) dy \text{ for } x \in \Omega_\varepsilon.$$

Proposition 3.13. If $u : \Omega \rightarrow \mathbb{R}$ be a locally integrable function, then

- (i) $u^\varepsilon \in C_c^\infty(\Omega_\varepsilon)$.
- (ii) $u^\varepsilon \rightarrow u$ a.e. as $\varepsilon \rightarrow 0^+$.
- (iii) If $u \in C(\Omega)$, then $u^\varepsilon \rightarrow u$ uniformly on compact subsets of Ω .
- (iv) If $1 \leq p < +\infty$ and $u \in L_{loc}^p(\Omega)$, then $u^\varepsilon \rightarrow u$ in $L_{loc}^p(\Omega)$.

Theorem 3.14 (uniform continuity in time). Let the initial condition $u_0 \in C(\mathbb{R}^n) \cap W^{1, \infty}(\mathbb{R}^n)$ be bounded, periodic and satisfy $\|Du_0\|_{L^\infty(\mathbb{R}^n)} \leq E_0$ (E_0 is a positive constant). If u is the viscosity solution of Problem (3.2) in $\mathbb{R}^n \times [0, T)$, then u is uniformly continuous in time.

The proof of Theorem 3.14 is given in Appendix.

To discretize the parabolic equation, we adopt a central difference scheme for the diffusion term and an upwind scheme for the convection term. Here and after, we choose the Finsler norm to be $H(\xi) = (|\xi_1|^p + |\xi_2|^p)^{1/p}$, where $\xi = (\xi_1, \xi_2)^T$, $1 < p < +\infty$.

According to [45], for $x = (x_1, x_2) \in \mathbb{R}^2$, we obtain

$$\begin{aligned} u_{i,j}^{n+1} &= u_{i,j}^n + \Delta t \cdot m_{i,j} (D^{x_1} V_1 + D^{x_2} V_2) \\ &\quad - \Delta t \left[(q_1)_{i,j} H(Du_{i,j}^n - w_{i,j}) \frac{\partial H}{\partial \xi_1} (Du_{i,j}^n - w_{i,j}) + (q_2)_{i,j} H(Du_{i,j}^n - w_{i,j}) \frac{\partial H}{\partial \xi_2} (Du_{i,j}^n - w_{i,j}) \right] \\ &\quad + \mu \Delta t \left[\left(\frac{\partial H}{\partial \xi_1} (Du_{i,j}^n) \right)^2 D^{x_1 x_1} u_{i,j}^n + 2 \frac{\partial H}{\partial \xi_1} (Du_{i,j}^n) \frac{\partial H}{\partial \xi_2} (Du_{i,j}^n) D^{x_1 x_2} u_{i,j}^n + \left(\frac{\partial H}{\partial \xi_2} (Du_{i,j}^n) \right)^2 D^{x_2 x_2} u_{i,j}^n \right], \end{aligned}$$

where

$$\begin{aligned} H(Du_{i,j}^n - w_{i,j}) &= (|D^{x_1} u_{i,j}^n - (w_1)_{i,j}|^q + |D^{x_2} u_{i,j}^n - (w_2)_{i,j}|^q)^{1/q}, \quad \frac{\partial H}{\partial \xi_1} (Du_{i,j}^n) = |D^{x_1} u_{i,j}^n|^{q-1} \\ &\quad / (|D^{x_1} u_{i,j}^n|^q + |D^{x_2} u_{i,j}^n|^q)^{(q-1)/q}, \quad \frac{\partial H}{\partial \xi_2} (Du_{i,j}^n) = |D^{x_2} u_{i,j}^n|^{q-1} / (|D^{x_1} u_{i,j}^n|^q + |D^{x_2} u_{i,j}^n|^q)^{(q-1)/q}, \quad D^{x_1} V_1 = \\ &\quad \left[H(Du_{i+1,j}^n - w_{i+1,j}) \frac{\partial H}{\partial \xi_1} (Du_{i+1,j}^n - w_{i,j}) - H(Du_{i-1,j}^n - w_{i-1,j}) \frac{\partial H}{\partial \xi_1} (Du_{i-1,j}^n - w_{i,j}) \right] / (2\Delta x_1), \quad D^{x_2} V_2 = \\ &\quad \left[H(Du_{i,j+1}^n - w_{i,j+1}) \frac{\partial H}{\partial \xi_2} (Du_{i,j+1}^n - w_{i,j}) - H(Du_{i,j-1}^n - w_{i,j-1}) \frac{\partial H}{\partial \xi_2} (Du_{i,j-1}^n - w_{i,j}) \right] / (2\Delta x_2) \text{ and } q(x) = \\ &\quad (q_1(x), q_2(x)). \end{aligned}$$

According to the criteria of Barles and Souganidis [8], it is easy to prove that the approximate solution converges to a function u . Then, by Theorems 3.8 and 3.9, we conclude that u is the unique viscosity solution of the parabolic Finsler infinity Laplacian equation.

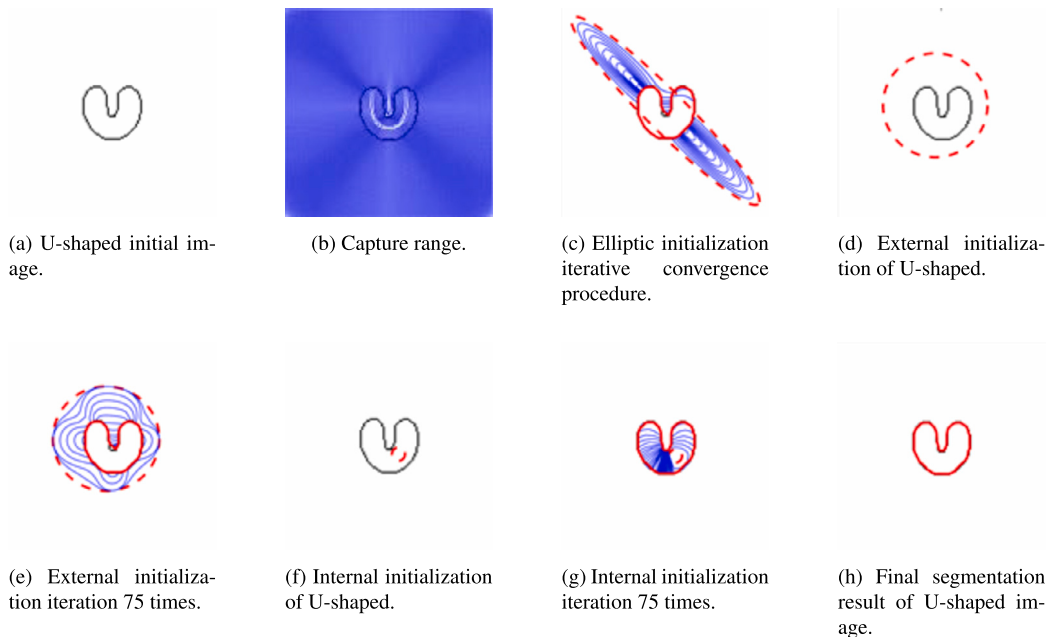


Fig. 1. The experimental results of the proposed model ($\mu = 0.1$) on U-shaped images. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

4. Experimental results

In this section, we present experimental results validating the segmentation performance of our proposed model. We demonstrate its expected performance characteristics, including a larger capture range and insensitivity to initial contour. Our model also effectively handles the case of edge discontinuities, concavity and sharp curvatures. Moreover, our model exhibits strong robustness against Gaussian noise, salt-and-pepper noise and speckle noise. Experimental results show that our model outperforms other comparative models in edge preservation and noise robustness. We also present performance metrics, including segmentation accuracy (Dice coefficient, Jaccard index and Recall rate), number of iterations and the time required to generate external force field.

Before performing segmentation, we normalize the image intensity to the range $[0, 1]$. We set $p = 4$, $\alpha = 0.1$ and $\beta = 0.06$ for the contours of all models with a time step $\Delta t = 0.5$.

Fig. 1 presents the capture range of our proposed model on U-shaped images and the convergence results under different initialization contours. The red dashed line represents the initial contour, the blue solid line indicates the iterative process, while the red solid line signifies the final convergence result. The results in Fig. 1(b) demonstrate that our model extrapolates the gradient vector field, thereby increasing the capture range. In Fig. 2, we present three types of images to illustrate our model's robustness in addressing edge discontinuities, concavities and sharp curvatures. These results demonstrate an outstanding performance, further confirming the effectiveness of our model in handling edges.

In actual image segmentation, images are often affected by noise. To verify the robustness of our proposed model ($\mu = 0.1$) with noise, we test the model on images with Gaussian noise, speckle noise and salt-and-pepper noise. We compare with the GGVF model ($\mu = 0.2, k = 0.5$), ILGVF model ($\mu = 0.1$), ADF model ($\sigma = 1, k = 0.05$) and GVFOM ($\mu = 0.2, \lambda = 11$). The Figs. 3 shows the segmentation results of each model after sixty iterations on the images with the three types of noise. Evidently, our model exhibits good convergence across all three noisy images and performs particularly well in the case of salt-and-pepper noise. In contrast, the GGVF and the ILGVF model are adversely affected by the noise points, leading to degraded segmentation results.

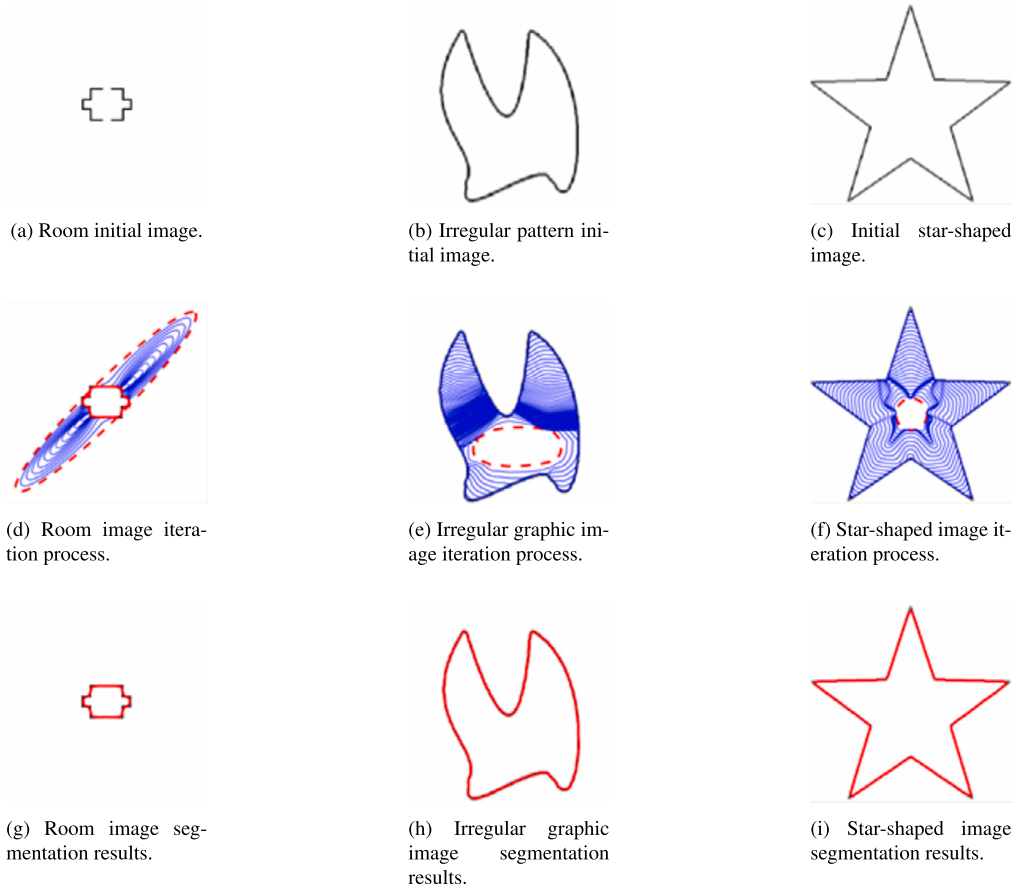


Fig. 2. The experimental results of the proposed model ($\mu = 0.1$) on the Room image, irregular shape image and star-shaped image.

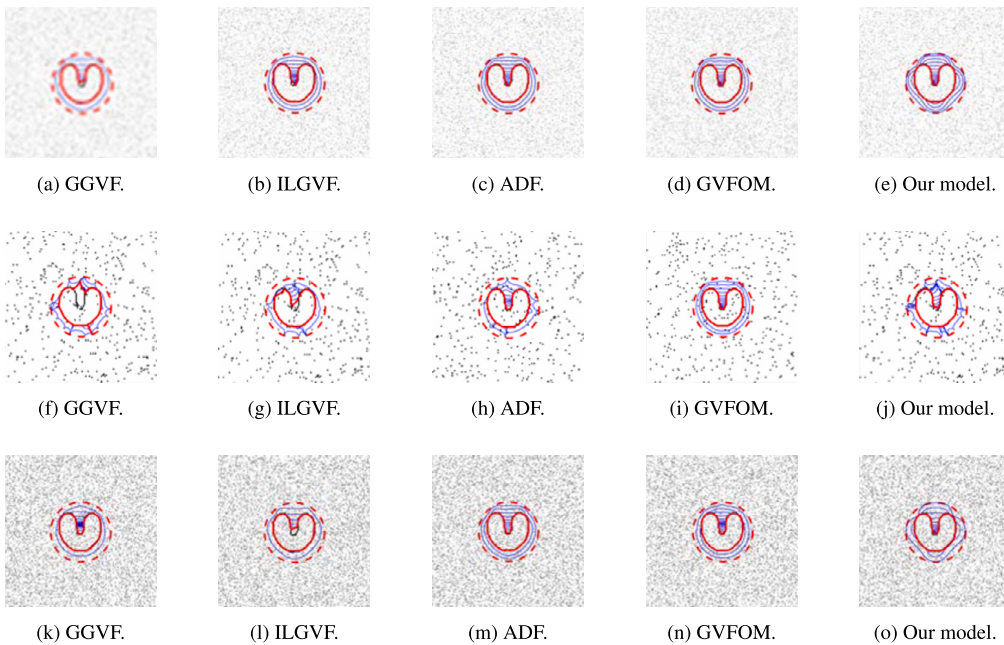


Fig. 3. Test results on different noisy images. From top to bottom, speckle noise 0.001, salt-and-pepper noise 0.05 and Gaussian noise 0.06. The first column: GGVF ($\mu = 0.2, k = 0.5$). The second column: ILGVF ($\mu = 0.1$). The third column: ADF model ($\sigma = 1, k = 0.05$). The fourth column: GVFOM ($\mu = 0.2, \lambda = 11$). The fifth column: our model ($\mu = 0.1$).

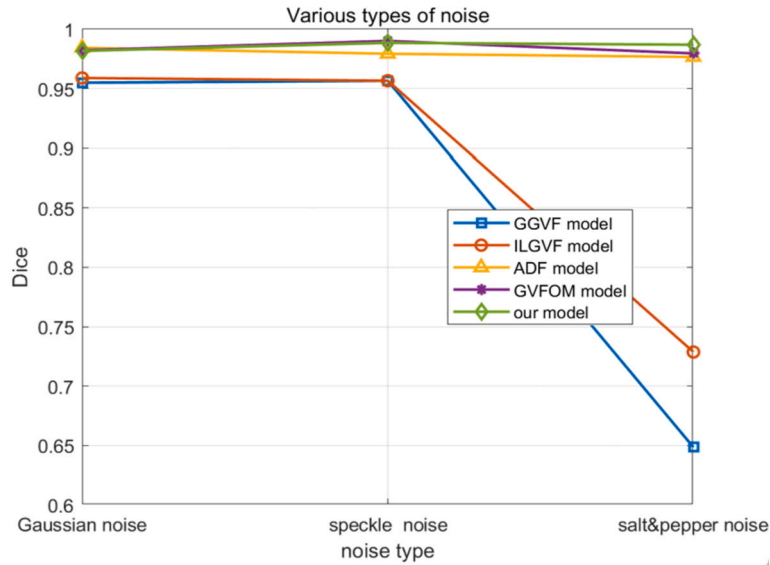


Fig. 4. The Dice coefficient at different noise.

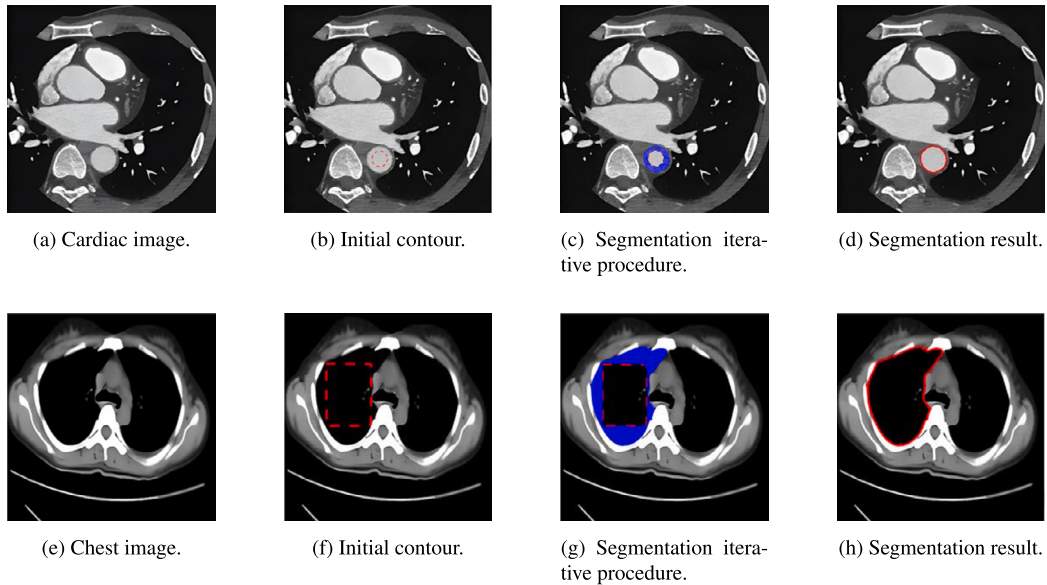


Fig. 5. The segmentation results of our proposed method ($\mu = 0.1$) on cardiac and chest images.

To further validate the robustness of our model against Gaussian noise, speckle noise and salt-and-pepper noise, we present the performance of the model's Dice coefficient under different noise conditions Fig. 4. Clearly, our model has higher Dice coefficient under three types of noise conditions than other models. It is intuitively clear that our model can still maintain excellent segmentation performance when confronted with these three types of noise interference.

Additionally, we test our model on cardiac and chest CT images. In Fig. 5, the red dashed line represents the initial contour, the blue area indicates the boundary changes during the iterative convergence process and the red solid line shows the final segmentation result. From Fig. 5(g) and 5(h), it is evident that our model maintains a strong ability to preserve details in medical images which require precise segmentation.

Fig. 6 shows that the GVF model performs poorly, while the GGVF model, the NGVF model, the ILGVF model and the ADF model achieve complete segmentation. However, these methods still have

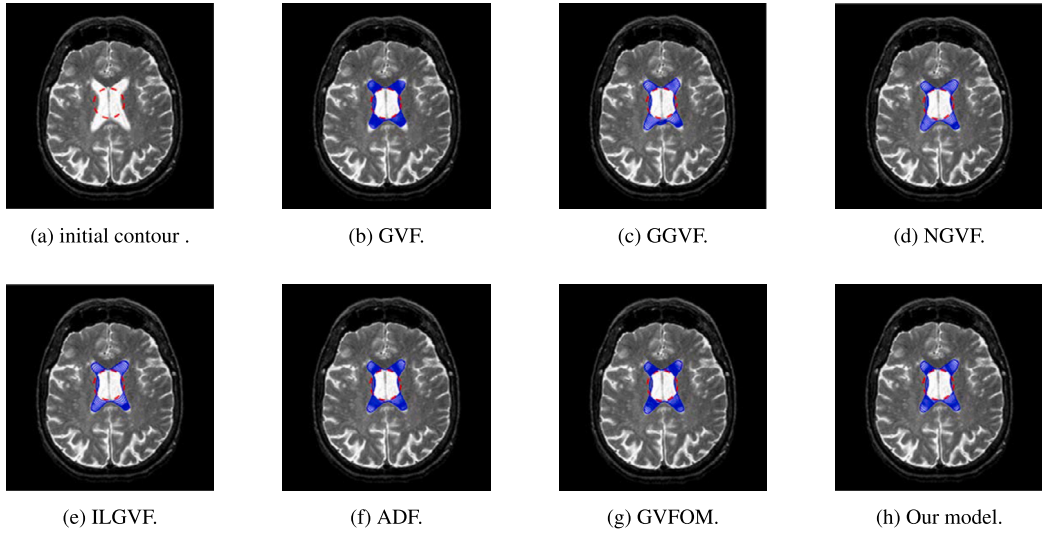


Fig. 6. Example on cortex image. (a) initial contour. (b) experimental results and evolution of GVF model ($\mu=0.2$). (c) experimental results and evolution of GGVF model ($\mu = 0.2, k = 0.05$). (d) experimental results and evolution of NGVF model ($\mu = 0.2$) (e) experimental results and evolution of ILGVF model ($\mu = 0.2$). (f) experimental results and evolution of ADF model ($\sigma = 1.5, k = 0.05$). (g) experimental results and evolution of GVFOM model ($\mu = 0.1, \lambda = 135$). (h) experimental results and evolution of our model ($\mu = 0.2$). The blue solid line indicates the iterative process and the red solid line signifies the final convergence result.

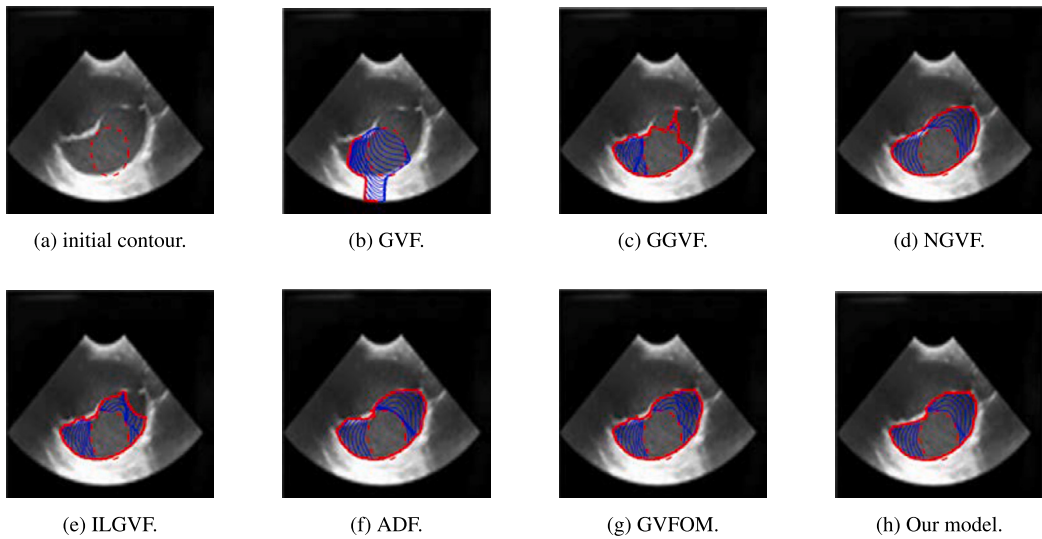


Fig. 7. Example on ultrasound image. (a) initial contour. (b) experimental results and evolution of GVF model ($\mu=0.2$). (c) experimental results and evolution of GGVF model ($\mu = 0.2, k = 0.05$). (d) experimental results and evolution of NGVF model ($\mu = 0.2$) (e) experimental results and evolution of ILGVF model ($\mu = 0.1$). (f) experimental results and evolution of ADF model ($\sigma = 0.5, k = 0.05$). (g) experimental results and evolution of GVFOM model ($\mu = 0.1, \lambda = 120$). (h) experimental results and evolution of our model ($\mu = 0.1$). The blue solid line indicates the iterative process and the red solid line signifies the final convergence result.

limitations in edge detection, particularly at the lower right corner of the cortex region. In contrast, our model demonstrates superior performance, effectively capturing the edges of the shapes to be segmented. In Fig. 7, Fig. 8 and Fig. 9, we further test the segmentation performance of our model under weak edges and noise interference conditions. In Fig. 7, the GVF, GGVF and ILGVF models are affected by noise and weak edges. Our model is as accurate as the ADF and GVFOM models. It also shows better performance in edge details. In Fig. 8, the goal is to extract a peach with a blurred edge. This causes other models to be attracted by adjacent edges, which can lead to leakage near weak boundaries. Experimental results show

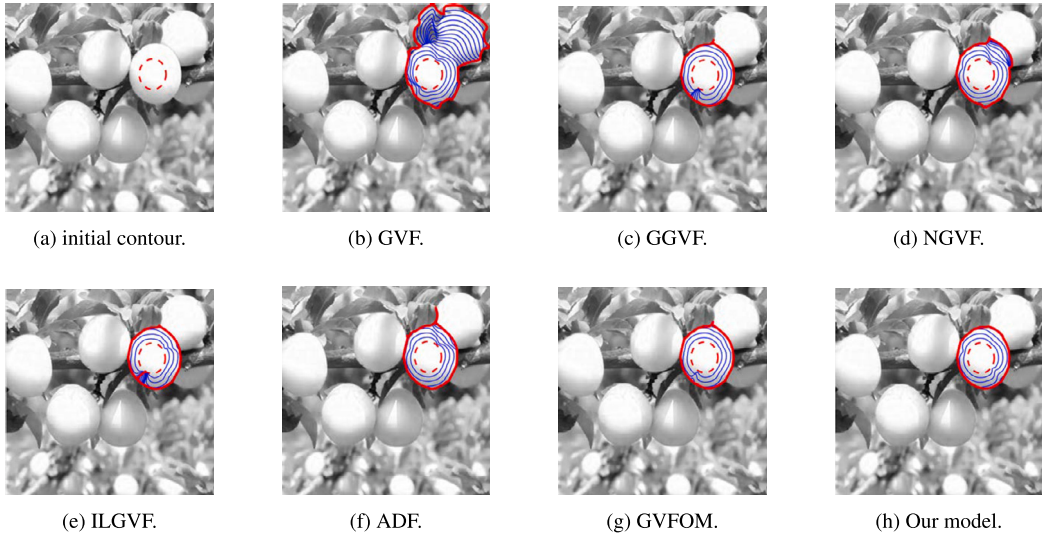


Fig. 8. Example on fruit image. (a) initial contour. (b) experimental results and evolution of GVF model ($\mu=0.2$). (c) experimental results and evolution of GGVF model ($\mu = 0.2, k = 0.05$). (d) experimental results and evolution of NGVF model ($\mu = 0.2$) (e) experimental results and evolution of ILGVF model ($\mu = 0.1$). (f) experimental results and evolution of ADF model ($\sigma = 1, k = 0.05$). (g) experimental results and evolution of GVFOM model ($\mu = 0.1, \lambda = 70$). (h) experimental results and evolution of our model ($\mu = 0.1$). The blue solid line indicates the iterative process and the red solid line signifies the final convergence result.

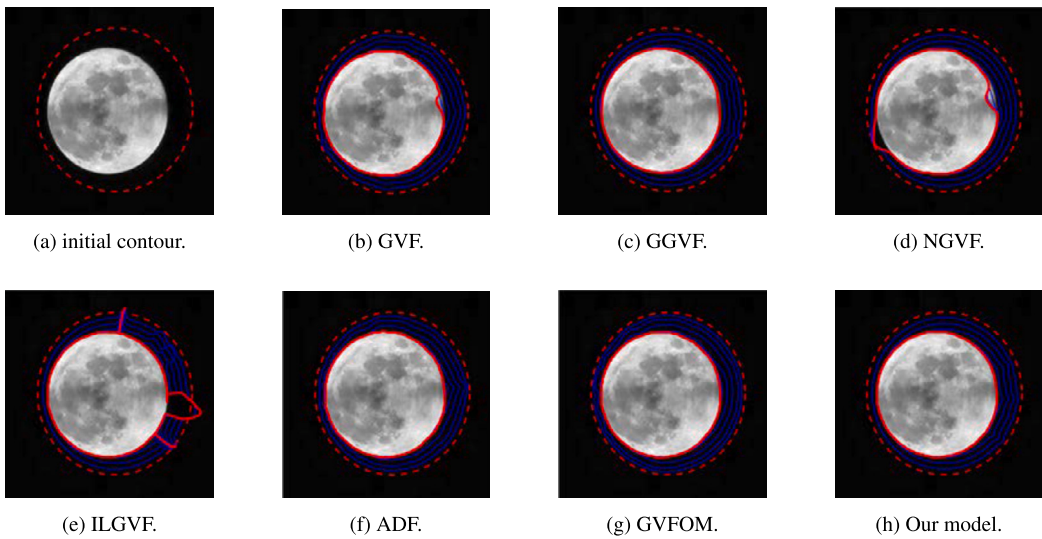


Fig. 9. Example on moon image. (a) initial contour. (b) experimental results and evolution of GVF model ($\mu = 0.2$). (c) experimental results and evolution of GGVF model ($\mu = 0.2, k = 0.05$). (d) experimental results and evolution of NGVF model ($\mu = 0.2$) (e) experimental results and evolution of ILGVF model ($\mu = 0.1$). (f) experimental results and evolution of ADF model ($\sigma = 1.5, k = 0.05$). (g) experimental results and evolution of GVFOM model ($\mu = 0.1, \lambda = 200$). (h) experimental results and evolution of our model ($\mu = 0.1$). The blue solid line indicates the iterative process and the red solid line signifies the final convergence result.

that our model accurately extracts the peach, successfully preserve the weak edges and outperforming other models. As shown in Fig. 9, there is a structurally weak region on the right side of the Moon, accompanied by noise around the edges. The GVF, GGVF, and ILGVF models are prone to being disturbed by weak edges, which can lead to incorrect convergence results. In contrast, our model successfully converges while effectively suppressing noise interference and preserving weak edges.

Table 1 presents the computation time and the number of iterations required for generating external force fields, including the GVF, GGVF, NGVF, ILGVF, ADF, GVFOM and our model. As shown, the computation time of the proposed method is comparable to that of the GVF model. To ensure fairness

Table 1
CPU Time (s) of the External Force Generation for Seven Models.

Image	U-shape	Cortex	Ultrasound	Fruit	Moon
Size	128×128	243×241	160×160	280×280	178×176
Iterations	75	45	75	80	60
GVF	0.207	6.532	4.468	8.621	4.811
GGVF	0.391	6.442	5.558	9.617	5.276
NGVF	0.198	4.522	4.898	7.470	3.849
ILGVF	0.362	7.234	5.863	17.694	8.001
ADF	0.201	4.157	4.015	8.663	5.640
GVFOM	0.464	12.605	7.901	20.384	12.194
Proposed	0.625	9.456	5.739	21.604	5.747

Table 2
The Dice coefficients of each model.

Image	Cortex image	Ultrasound	Fruit	Moon
GVF	0.9524	0.1679	0.5926	0.8710
GGVF	0.9267	0.6555	0.9626	0.9840
NGVF	0.9431	0.9423	0.9088	0.8409
ILGVF	0.9232	0.8578	0.9811	0.7626
ADF	0.9500	0.9390	0.9455	0.9352
GVFOM	0.9467	0.9529	0.9655	0.9851
Proposed	0.9522	0.9716	0.9819	0.9694

Table 3
The Jaccard index of each model.

Image	Cortex image	Ultrasound	Fruit	Moon
GVF	0.8943	0.1352	0.4210	0.8526
GGVF	0.9015	0.7474	0.9278	0.9678
NGVF	0.8927	0.9509	0.8328	0.8127
ILGVF	0.8876	0.7510	0.9583	0.7278
ADF	0.9133	0.8850	0.8967	0.9143
GVFOM	0.9065	0.8925	0.9615	0.9428
Proposed	0.9137	0.9481	0.9640	0.9685

Table 4
The Recall rate of each model.

Image	Cortex image	Ultrasound	Fruit	Moon
GVF	0.9305	0.1354	0.4632	0.8427
GGVF	0.8583	0.7856	0.9308	0.9596
NGVF	0.9112	0.9747	0.8336	0.8712
ILGVF	0.9403	0.7814	0.9523	0.8208
ADF	0.9332	0.9703	0.9671	0.9332
GVFOM	0.9260	0.9750	0.9518	0.9518
Proposed	0.9335	0.9755	0.9751	0.9688

and objectivity, each model underwent the same number of serpentine curve evolutions and force field iterations.

A quantitative analysis of segmentation performance is conducted using the Dice coefficient, Jaccard index and Recall rate, with the results from four test images presented in Table 2 to 4. Statistical analysis shows that the proposed model has achieved better values in all evaluation metrics in the majority of cases. This result confirms that our model has obvious advantages in segmentation accuracy and its performance is more stable. Additional segmentation results on actual images are presented in Fig. 10. In Fig. 10, the red dashed line denotes the initial contour, the blue solid line represents the trajectory of the evolving curve during iteration and the red solid line indicates the final segmentation boundary. In subfigures (a) (c) (d), (g) and (h), the initial contour is located outside the target object. Through iterative convergence, the evolving curve progressively approaches the object boundary, achieving good segmentation results while

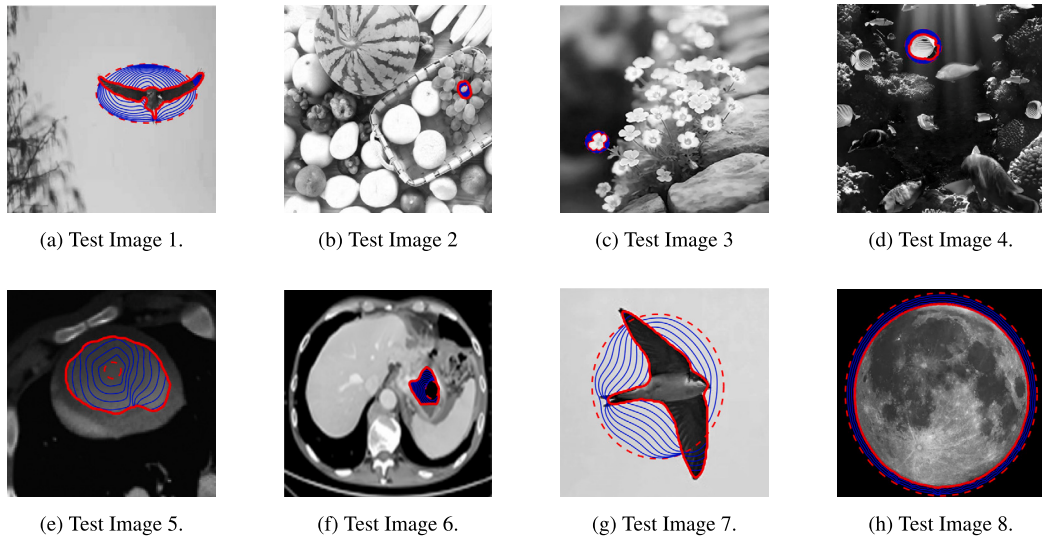


Fig. 10. Test results on actual images.

remaining unaffected by noise. In subfigures (b), (e) and (f), the initial contour is positioned inside the target object. Despite being close to the contours of surrounding regions, the segmentation process is not influenced by nearby edges and successfully converges to the object boundaries. These results demonstrate that the proposed model effectively expands the capture range of the force field, reduces sensitivity to the initial contour and achieves segmentation of target regions while remaining robust to noise.

To assess the convergence of energy functional, we conduct energy convergence experiments on the test images. As can be seen from Fig. 11, the values of the energy functional gradually decrease with the increase of the number of iterations on different types of images, without showing obvious fluctuations or divergence. After about 20 to 30 iterations, it tends to being stable and eventually achieving convergence.

5. Conclusion

In this paper, we propose a new improved active contour model under Finsler norm. The energy functional of our model includes the L^∞ -norm regularization term and the data fitting term under the Finsler norm, which is closely related to Finsler ∞ -Laplacian. We establish the comparison principle of Problem (3.2) using the method of double variables and demonstrate the existence of the viscosity solution u using the Perron method. Additionally, we prove the Lipschitz continuity of u in space using the method of the continuous modulus and the uniform continuity of u in time using the mollifier argument. It is noteworthy that these results remain valid for $n \geq 2$. With the existence, uniqueness and regularity of the viscosity solution u of Problem (3.2), we then conduct a series of experiments to test the performance of our proposed model. We verify that our model is insensitive to initialization, has a large capture range and effectively addresses the case of edge discontinuities, concavity and sharp curvatures. Through comparative experiments, we demonstrate that our model exhibits good robustness with Gaussian noise, speckle noise and salt-and-pepper noise. Through comparative experiments, the proposed method demonstrated excellent performance in evaluation metrics such as visual effect, Dice coefficient, Jaccard index and Recall rate, showcasing a significant advantage in the preservation of edge. Additionally, we also provided the experimental verification results for model convergence. Compared with Deep Learning Methods, our work has its own advantages: Firstly, our method does not need large-scale labeled data. Secondly, our work provides detailed proofs of the existence, uniqueness and convergence related to the anisotropic model. Thirdly, our work shows stable segmentation performance due to avoiding the possible distribution shift between the training and test sets. In the future, we will integrate deep learning methods with our model.

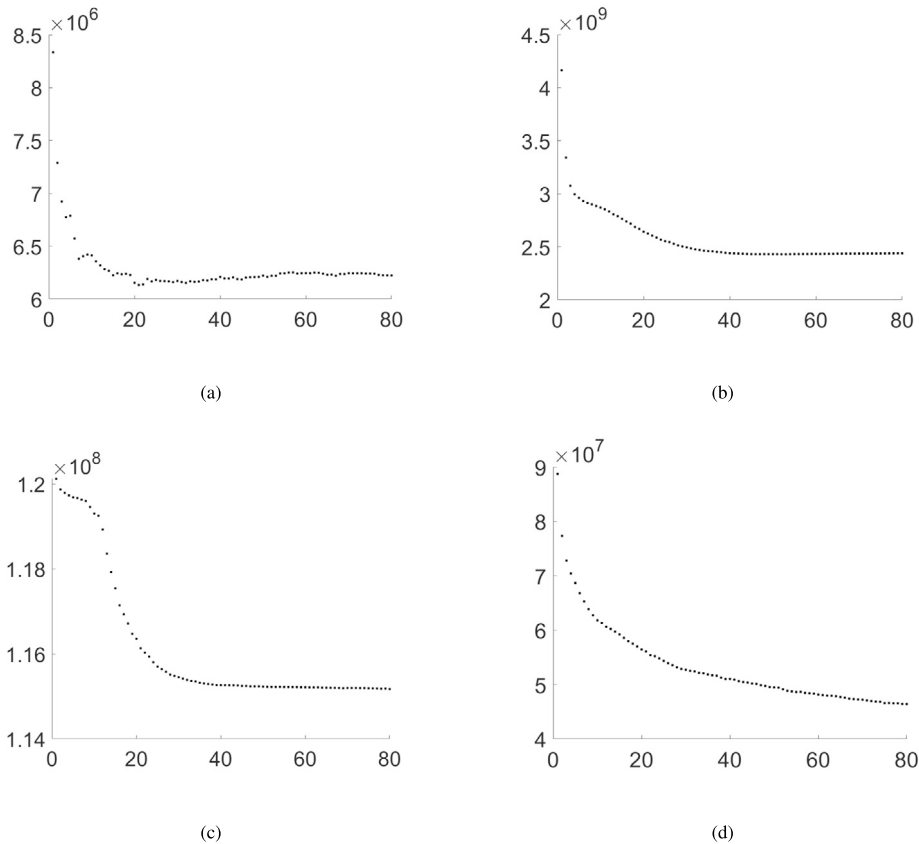


Fig. 11. The converge experiments of our model. (a) U-shaped image. (b) Chest image. (c) Fruit image. (d) Moon image.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the anonymous referees for their careful reading of the manuscript and useful suggestions and comments.

Appendix A

We present the proofs of several theorems in this section. Due to the high degeneracy of the Finsler ∞ -Laplacian, we need to provide proofs within the framework of the viscosity solution.

Proof of Theorem 3.8. Let $\Phi(x, 0)$ denote the initial condition of Problem (3.2) in $\mathbb{R}^n \times [0, T)$. For any $\lambda > 0$, we replace u with $u - \frac{\lambda}{T-t}$. Obviously, u is a strict subsolution of Problem (3.2) in $\mathbb{R}^n \times [0, T)$. Since $u \leq v$ follows from $u - \frac{\lambda}{T-t} \leq v$ in the limit $\lambda \rightarrow 0^+$. Therefore, we only need to prove

$$u_t + F^+(x, Du, D^2u) \leq -\frac{\lambda}{(T-t)^2} \leq -\frac{\lambda}{T^2}, \quad (5.1)$$

and

$$\lim_{t \rightarrow T^-} u(x, t) = -\infty. \quad (5.2)$$

Letting $M := \sup_{\mathbb{R}^n \times [0, T)} (u(x, t) - v(x, t))$, we want to show $M \leq 0$. To this end, we assume for contradiction that $M > 0$. Define

$$M_0 := \sup_{\mathbb{R}^n \times \mathbb{R}^n \times [0, T)} \{u(x, t) - v(y, t) - (4\varepsilon)^{-1} H^*(x - y)^4\}, \quad \varepsilon > 0.$$

Let

$$\varphi(x, y, t) := (4\varepsilon)^{-1} H^*(x - y)^4.$$

Due to the boundedness of u and $-v$, the function $d(x, y, t) := u(x, t) - v(y, t) - \varphi(x, y, t)$ satisfies $d(x + 2h, y + 2h, t) = d(x, y, t)$, for any $h \in \mathbb{Z}^n$. Combined with (5.2), there exists a maximum point $(x_0, y_0, t_0) \in \mathbb{R}^n \times \mathbb{R}^n \times [0, T)$ at which the supremum of M_0 is achieved. Consequently, we have

$$M_0 = u(x_0, t_0) - v(y_0, t_0) - \varphi(x_0, y_0, t_0) \geq M > 0.$$

We claim $t_0 > 0$. Assume for contradiction that $t_0 = 0$. Since u is a viscosity subsolution and v is a viscosity supersolution of Problem (3.2) in $\mathbb{R}^n \times [0, T)$, we have

$$u(x, 0) \leq \Phi(x, 0) \quad \text{for all } x \in \mathbb{R}^n,$$

and

$$v(x, 0) \geq \Phi(x, 0) \quad \text{for all } x \in \mathbb{R}^n.$$

Thus,

$$\begin{aligned} 0 < M &\leq M_0 = u(x_0, 0) - v(y_0, 0) - \varphi(x_0, y_0, t_0) \\ &\leq \Phi(x_0, 0) - \Phi(y_0, 0) - \varphi(x_0, y_0, t_0) \\ &\leq \sup_{\mathbb{R}^n \times \mathbb{R}^n} (\Phi(x, 0) - \Phi(y, 0) - \varphi(x_0, y_0, t_0)) \\ &= \Phi(x_1, 0) - \Phi(y_1, 0) - \varphi(x_1, y_1, t_1), \end{aligned}$$

where (x_1, y_1) is the point at which the supremum is attained. Letting $L := \|\Phi(x, 0)\|_{L^\infty(\mathbb{R}^n)}$, we have

$$0 \leq \Phi(x_1, 0) - \Phi(y_1, 0) - \varphi(x_1, y_1, t_1) \leq 2L - \varphi(x_1, y_1, t_1),$$

which implies

$$H^*(x_1 - y_1) \leq (8\varepsilon L)^{\frac{1}{4}}. \quad (5.3)$$

Then, due to the Lipschitz continuity of Φ , one obtains

$$0 \leq \Phi(x_1, 0) - \Phi(y_1, 0) - \varphi(x_1, y_1, t_1) \leq L_\Phi H^*(x_1 - y_1) \leq L_\Phi (8\varepsilon L)^{\frac{1}{4}}, \quad (5.4)$$

where L_Φ is the Lipschitz constant of Φ . Taking $\varepsilon \rightarrow 0^+$ in (5.4), we obtain

$$\lim_{\varepsilon \rightarrow 0^+} \Phi(x_1, 0) - \Phi(y_1, 0) - \varphi(x_1, y_1, t_1) = 0,$$

which contradicts the hypothesis. Consequently, for any sufficiently small $\varepsilon > 0$, there holds $t_0 > 0$.

By [16], for every $\varepsilon > 0$ and $\varphi(x, y, t)$, there exist $\tau_1, \tau_2 \in \mathbb{R}$ and $X, Y \in S^n$, such that

$$\begin{aligned} (\tau_1, p_1, X) &\in \overline{\mathcal{P}}^{2,+} u(x_0, t_0), (\tau_2, p_2, Y) \in \overline{\mathcal{P}}^{2,+} v(y_0, t_0), \\ -\left(\frac{1}{\gamma} + \|A\|\right) I &\leq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq A + \gamma A^2, \end{aligned} \quad (5.5)$$

and

$$\tau_1 - \tau_2 = \partial_t \varphi(x_0, y_0, t_0) = 0,$$

where $p_1 = \partial_x \varphi(x_0, y_0, t_0)$, $p_2 = -\partial_y \varphi(x_0, y_0, t_0)$ and $A = D^2 \varphi(x_0, y_0, t_0)$. By direct calculations, we obtain

$$p_1 = \partial_x \varphi(x_0, y_0, t_0) = \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) = p_2,$$

and

$$A = \varepsilon^{-1} H^*(x_0 - y_0)^2 \begin{pmatrix} B & -B \\ -B & B \end{pmatrix},$$

where $B = 3D_\xi H^*(x_0 - y_0) \otimes D_\xi H^*(x_0 - y_0) + H^*(x_0 - y_0) D_\xi^2 H^*(x_0 - y_0)$. Since u is a viscosity subsolution and v is a viscosity supersolution of Problem (3.2), we have

$$\begin{aligned} \tau_1 + b(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) &+ \\ R(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), X) &+ \\ Q(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), X) &\leq -\frac{\lambda}{T^2}, \end{aligned} \quad (5.6)$$

and

$$\begin{aligned} \tau_2 + b(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) &+ \\ R(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) &+ \\ + Q(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) &\geq 0. \end{aligned} \quad (5.7)$$

Now we claim $x_0 \neq y_0$. Otherwise, if $x_0 = y_0$, we obtain $p_1 = p_2 = 0$ and $A = 0$. By (5.5), as $\gamma \rightarrow 0^+$, we conclude $X = Y = 0$. Consequently, we have

$$Q^*(0, 0) = Q_*(0, 0), \quad R(x_0, 0, 0) = R(y_0, 0, 0) \text{ and } b(x_0, 0) = b(y_0, 0).$$

By (5.6) and (5.7), we have

$$\tau_1 + b(x_0, 0) + R(x_0, 0, 0) \leq -\frac{\lambda}{T^2}, \quad (5.8)$$

and

$$\tau_2 + b(y_0, 0) + R(y_0, 0, 0) \geq 0. \quad (5.9)$$

Subtracting (5.9) from (5.8) yields,

$$-\frac{\lambda}{T^2} \geq 0,$$

which contradicts the assumption $\lambda > 0$.

Since $x_0 \neq y_0$ and (5.5), choosing $\gamma = \frac{\varepsilon}{2H^*(x_0 - y_0)^2}$, we obtain

$$\begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq \frac{1}{\varepsilon} H^*(x_0 - y_0)^2 \begin{pmatrix} V & -V \\ -V & V \end{pmatrix}, \quad (5.10)$$

$$\begin{aligned} \text{where } V = & 3D_\xi H^*(x_0 - y_0) \otimes D_\xi H^*(x_0 - y_0) + H^*(x_0 - y_0) D_\xi^2 H^*(x_0 - y_0) + \\ & 9|D_\xi H^*(x_0 - y_0)|^2 D_\xi H^*(x_0 - y_0) \otimes D_\xi H^*(x_0 - y_0) + 6H^*(x_0 - y_0) \\ & D_\xi H^*(x_0 - y_0) \otimes D_\xi H^*(x_0 - y_0) D_\xi^2 H^*(x_0 - y_0) + H^*(x_0 - y_0)^2 \\ & D_\xi^2 H^*(x_0 - y_0)^2. \end{aligned}$$

Subtracting (5.7) from (5.6) and using the local Lipschitz continuity of the function b , we have

$$\begin{aligned} \frac{\lambda}{T^2} \leq & b(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) - b(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) \\ & + R(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) - R(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 \\ & D_\xi H^*(x_0 - y_0), X) + Q(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) \\ & - Q(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), X). \end{aligned} \quad (5.11)$$

By (5.10), we deduce $X \leq Y$. Due to the ellipticity of the function Q , it follows that

$$Q(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) - Q(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), X) \leq 0. \quad (5.12)$$

For $t = t_0$ and $p_1 = p_2 = \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)$, it can be verified that there exist constants C_0 , C_1 and C_2 such that the following inequalities hold

$$\begin{aligned} H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w) & \leq C_0, \\ |D_\xi H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w)| & \leq C_1, \end{aligned}$$

and

$$|D_\xi^2 H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w)| \leq C_2.$$

Therefore,

$$\begin{aligned} & R(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) \\ & - R(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), X) \\ & = m(x_0) \operatorname{div} \left[H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w) \right. \\ & \quad \left. \times D_\xi H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w) \right] \\ & - m(y_0) \operatorname{div} \left[H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w) \right. \end{aligned}$$

$$\begin{aligned}
& \times D_\xi H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) \Big] \\
& = m(x_0) \operatorname{trace} \left(D_\xi H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) \right. \\
& \quad \otimes D_\xi H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) (X - Dw) \\
& \quad + H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) \\
& \quad \times D_\xi^2 H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) (X - Dw) \Big) \\
& \quad - m(y_0) \operatorname{trace} \left(D_\xi H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) \right. \\
& \quad \otimes D_\xi H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) (Y - Dw) \\
& \quad + H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) \cdot \\
& \quad \left. D_\xi^2 H \left(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w \right) (Y - Dw) \right) \\
& \leq C \operatorname{trace} \begin{pmatrix} m(x_0)(X - Dw) & 0 \\ 0 & -m(y_0)(Y - Dw) \end{pmatrix} \\
& = C \operatorname{trace} \left(N \begin{pmatrix} X - Dw & 0 \\ 0 & -(Y + Dw) \end{pmatrix} \right) \\
& = C \operatorname{trace} \left(\eta^T \begin{pmatrix} X - Dw & 0 \\ 0 & -(Y + Dw) \end{pmatrix} \eta \right) \\
& = C \sum_{i=1}^4 \eta_i^T \begin{pmatrix} X - Dw & 0 \\ 0 & -(Y + Dw) \end{pmatrix} \eta_i \\
& \leq \sum_{i=1}^4 C_w \eta_i^T \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \eta_i,
\end{aligned} \tag{5.13}$$

where C is a constant related to H , $D_\xi H$, $D_\xi^2 H$, C_w is a constant related to w , H , $D_\xi H$, $D_\xi^2 H$, $N :=$

$$\begin{pmatrix} \frac{m(x_0)I}{\sqrt{m(x_0)m(y_0)}I} & \frac{\sqrt{m(x_0)m(y_0)}I}{m(y_0)I} \end{pmatrix} = \eta \eta^T, \eta \text{ is a column vector and } \eta_i \text{ is the } i^{\text{th}} \text{ column of } \eta.$$

By (5.13), we obtain

$$\begin{aligned}
\sum_{i=1}^4 C_w \eta_i^T \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \eta_i & \leq \frac{1}{\varepsilon} H^*(x_0 - y_0)^2 C_w \sum_{i=1}^4 \eta_i^T \begin{pmatrix} V & -V \\ -V & V \end{pmatrix} \eta_i \\
& \leq \frac{1}{\varepsilon} H^*(x_0 - y_0)^2 C_w \operatorname{trace} \left(N \begin{pmatrix} V & V \\ -V & V \end{pmatrix} \right) \\
& \leq \frac{1}{\varepsilon} H^*(x_0 - y_0)^2 C_w \left(\sqrt{m(x_0)} - \sqrt{m(y_0)} \right)^2 \operatorname{trace}(V) \\
& \leq \frac{1}{\varepsilon} H^*(x_0 - y_0)^4 C_w k_m^2 \operatorname{trace}(V),
\end{aligned} \tag{5.14}$$

where k_m is the Lipschitz constant of the function $m^{\frac{1}{2}}$. Due to the boundedness of $\operatorname{trace}(V)$, by (5.13) and (5.14), one has

$$\begin{aligned}
& R(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), Y) - R(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0), X) \\
& \leq \frac{1}{\varepsilon} C H^*(x_0 - y_0)^4,
\end{aligned} \tag{5.15}$$

where $C = C(w, k_m, H, D_\xi H, D_\xi^2 H)$ is a constant. By Property P_3 , one has

$$\begin{aligned}
& b(x_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) \\
& - b(y_0, \varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) \\
& \geq -k_q |H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0))| \cdot \\
& D_\xi H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0)) |H^*(x_0 - y_0)|.
\end{aligned} \tag{5.16}$$

Thus, by (5.11), (5.12), (5.15) and (5.16), we have

$$\begin{aligned}
\lambda & \leq k_q \left| H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w) \right. \\
& D_\xi H(\varepsilon^{-1} H^*(x_0 - y_0)^3 D_\xi H^*(x_0 - y_0) - w) \left. \right| H^*(x_0 - y_0) T^2 \\
& + \left(\frac{1}{\varepsilon} C H^*(x_0 - y_0)^4 T^2 \right).
\end{aligned} \tag{5.17}$$

Let $\rho := \max(\|u(x, t)\|_{L^\infty(\mathbb{R}^n \times [0, T])}, \|v(x, t)\|_{L^\infty(\mathbb{R}^n \times [0, T])})$. Similarly to (5.3), we deduce

$$H^*(x_0 - y_0) \leq (8\rho\varepsilon)^{\frac{1}{4}}.$$

For any $l > 0$, we define $M_* := \lim_{l \rightarrow 0^+} \sup_{H^*(x-y) \leq l, t \in [0, T]} (u(x, t) - v(y, t))$ and $M_l := \sup_{H^*(x-y) \leq l} (u(x, t) - v(y, t))$.

Considering sequences (x_n^l, y_n^l, t_n^l) such that

$$u(x_n^l, t_n^l) - v(y_n^l, t_n^l) \geq M_l - \frac{1}{n},$$

we have

$$H^*(x_n^l - y_n^l) \leq l,$$

and

$$M_l - \frac{1}{n} - \frac{l^4}{4\varepsilon} \leq u(t_n^l, t_n^l) - v(y_n^l, t_n^l) - (4\varepsilon)^{-1} H^*(x_n^l - y_n^l)^4 \leq M_0 \leq u(x_0, t_0) - v(y_0, t_0).$$

Letting $l \rightarrow 0^+$, one has

$$\begin{aligned}
M_* - \frac{1}{n} & \leq u(x_0, t_0) - v(y_0, t_0) \\
& \leq \lim_{\varepsilon \rightarrow 0^+} (u(x_0, t_0) - v(y_0, t_0)) \\
& \leq \lim_{\varepsilon \rightarrow 0^+} \sup_{H^*(x-y) \leq (8\rho\varepsilon)^{\frac{1}{4}}, t \in [0, T]} (u(x, t) - v(y, t)) \\
& \leq \lim_{l \rightarrow 0^+} \sup_{H^*(x-y) \leq l, t \in [0, T]} (u(x, t) - v(y, t)) \\
& = M_*.
\end{aligned} \tag{5.18}$$

By (5.18), as $n \rightarrow +\infty$, it follows that

$$\lim_{\varepsilon \rightarrow 0^+} (u(x_0, t_0) - v(y_0, t_0)) = M_*.$$

Obviously, we have

$$\lim_{\varepsilon \rightarrow 0^+} M_0 = M_*.$$

Then, one has

$$\lim_{\varepsilon \rightarrow 0^+} (u(x_0, t_0) - v(y_0, t_0) - M_*) = \lim_{\varepsilon \rightarrow 0^+} \frac{H^*(x_0 - y_0)^4}{4\varepsilon} = 0.$$

As $\varepsilon \rightarrow 0^+$, we have $H^*(x_0 - y_0) = 0$, which contradicts the inequality (5.17).

Consequence, we have concluded $M \leq 0$, which implies $u \leq v$ in $\mathbb{R}^n \times [0, T]$. $\square$

Proof of Theorem 3.9. Construction begins with a viscosity subsolution U^- of Problem (3.2) in $\mathbb{R}^n \times [0, T]$. Let

$$U^- := \inf_{x \in \mathbb{R}^n} (u_0) - Ct,$$

where $C := \zeta_q C_0 C_1$, $\|q(x)\|_{L^\infty(\mathbb{R}^n)} = \zeta_q$ and C_0, C_1 are the constants defined as Lemma 3.7. The function U^- is twice differentiable in space variable, once differentiable in time variable and bounded with the same periodicity as u_0 . One can verify that U^- is a viscosity subsolution of Problem (3.2) in $\mathbb{R}^n \times [0, T]$. Similarly, we construct the viscosity supersolution $U^+ := \sup_{x \in \mathbb{R}^n} (u_0) + Ct$. Obviously, we have $U^-(x, 0) \leq U^+(x, 0)$.

Define

$$u(x, t) := \sup\{\omega(x, t) : \omega(x, t) \text{ is a subsolution of Problem (3.2), } U^- \leq \omega(x, t) \leq U^+ \\ \text{with the same periodicity as } u_0, (x, t) \in \mathbb{R}^n \times [0, T]\}.$$

By the Perron method, $u(x, t)$ is a periodic viscosity solution of Problem (3.2), keeping the same periodicity as the initial condition u_0 . Since U^- , U^+ are bounded, the viscosity solution u is also bounded. Let $u^*(x, t) := \limsup_{(y, r) \rightarrow (x, t)} u(y, r)$, which represents the USC-envelope of $u(x, t)$ in $\mathbb{R}^n \times [0, T]$. By [16], the $u^*(x, t)$ is a viscosity subsolution of Problem (3.2). Similarly, the LSC-envelope $u_*(x, t) := \liminf_{(y, r) \rightarrow (x, t)} u(y, r)$ is a viscosity supersolution of Problem (3.2). Consequently, $u^*(x, t) \geq u_*(x, t)$ holds for all $(x, t) \in \mathbb{R}^n \times [0, T]$.

Since $u^*(x, t)$ is a viscosity subsolution and $u_*(x, t)$ a supersolution of Problem (3.2) in $\mathbb{R}^n \times [0, T]$, Theorem 3.8 implies $u^*(x, t) \leq u_*(x, t)$. Combining this with $u^* \geq u_*$, it follows that $u^*(x, t) = u_*(x, t)$ holds for all $(x, t) \in \mathbb{R}^n \times [0, T]$. Consequently, $u(x, t)$ is continuous on $\mathbb{R}^n \times [0, T]$. $\square$

Proof of Theorem 3.10. To prove (3.4), it is sufficient to show that for small enough $\varepsilon > 0$,

$$u(x, t) - u(y, t) \leq \Phi^\varepsilon(x, y, t) := E(t)(H^*(x - y)^2 + \varepsilon^2)^{\frac{1}{2}}.$$

Define

$$\overline{M} := \sup_{(x, y, t) \in \mathbb{R}^n \times \mathbb{R}^n \times [0, T]} \{u(x, t) - u(y, t) - \Phi^\varepsilon(x, y, t)\}.$$

For the sake of contradiction, assume that $\overline{M} > 0$. Define

$$M' := \sup_{(x,y,t) \in \mathbb{R}^n \times \mathbb{R}^n \times [0,T)} \left\{ u(x,t) - u(y,t) - \Phi^\varepsilon(x,y,t) - \frac{\lambda}{T-t} \right\} > 0$$

for sufficiently small $\lambda > 0$. Let (x_1, y_1, t_1) be the maximum point of M' . Obviously, $x_1 \neq y_1$.

We claim $t_1 > 0$. Otherwise $t_1 = 0$, then

$$u_0(x_1) - u_0(y_1) - \Phi^\varepsilon(x_1, y_1, 0) > 0.$$

That is,

$$u_0(x_1) - u_0(y_1) > E_0(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}} > E_0 H^*(x_1 - y_1),$$

which leads to a contradiction.

Taking $\Phi(x, y, t) := \Phi^\varepsilon(x, y, t) + \frac{\lambda}{T-t}$, by [16], for each $\varepsilon > 0$, there exist $\tau_1, \tau_2 \in \mathbb{R}$ and $X, Y \in S^n$ such that

$$\begin{aligned} \tau_1 - \tau_2 &= D_t \Phi(x_1, y_1, t_1) = E'(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}} + \frac{\lambda}{(T - t_1)^2}, \\ (\tau_1, p_1, X) &\in \bar{p}^{2,+} u(x_1, t_1), (\tau_2, p_2, Y) \in \bar{p}^{2,-} v(y_1, t_1), \end{aligned}$$

and

$$-\left(\frac{1}{\varepsilon} + \|A\|\right) I \leq \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \leq A + \varepsilon A^2, \quad (5.19)$$

where $p_1 = \partial_x \Phi(x_1, y_1, t_1) = E(t_1) \frac{H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}$, $p_2 = -\partial_y \Phi(x_1, y_1, t_1) = p_1$ and $A = D^2 \Phi(x_1, y_1, t_1) = E(t_1) \begin{pmatrix} V & -V \\ -V & V \end{pmatrix}$ with

$$\begin{aligned} V &= ((H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} - H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{3}{2}}) D_\xi H^*(x_1 - y_1) \otimes \\ &\quad D_\xi H^*(x_1 - y_1) + H^*(x_1 - y_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} D_\xi^2 H^*(x_1 - y_1). \end{aligned}$$

Choosing $\varepsilon = \frac{1}{2E(t_1)} (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}} > 0$, we obtain

$$A + \varepsilon A^2 = E(t_1) \begin{pmatrix} U & -U \\ -U & U \end{pmatrix},$$

where

$$\begin{aligned} U &= \left((H^*(x_1 - y_1)^2 + \varepsilon^2) \right)^{-\frac{1}{2}} D_\xi H^*(x_1 - y_1) \otimes D_\xi H^*(x_1 - y_1) \\ &\quad - H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{3}{2}} D_\xi H^*(x_1 - y_1) \otimes D_\xi H^*(x_1 - y_1) \\ &\quad + H^*(x_1 - y_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} D_\xi^2 H^*(x_1 - y_1) \\ &\quad + (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} |D_\xi H^*(x_1 - y_1)|^2 \\ &\quad + H^*(x_1 - y_1)^4 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{5}{2}} |D_\xi H^*(x_1 - y_1)|^2 \end{aligned}$$

$$\begin{aligned}
& -2H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{3}{2}} |D_\xi H^*(x_1 - y_1)|^2 \\
& + H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} (D_\xi^2 H^*(x_1 - y_1))^2 \\
& + 2H^*(x_1 - y_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} D_\xi^2 H^*(x_1 - y_1) \cdot (D_\xi H^* \otimes D_\xi H^*) \\
& - 2H^*(x_1 - y_1)^3 (H^*(x_1 - y_1) + \varepsilon^2)^{-\frac{3}{2}} D_\xi^2 H^*(x_1 - y_1) \cdot (D_\xi H^* \otimes D_\xi H^*).
\end{aligned}$$

Since u is a viscosity subsolution and v is a viscosity supersolution of Problem (3.2), we have

$$\begin{aligned}
& \tau_1 + b \left(x_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}} \right) \\
& + R \left(x_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, X \right) \\
& + Q \left(\frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, X \right) \leq 0,
\end{aligned} \tag{5.20}$$

and

$$\begin{aligned}
& \tau_2 + b \left(y_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}} \right) \\
& + R \left(y_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, Y \right) \\
& + Q \left(\frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, Y \right) \geq 0.
\end{aligned} \tag{5.21}$$

Subtracting (5.21) from (5.20), we derive

$$\begin{aligned}
& \tau_1 - \tau_2 + b \left(x_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}} \right) \\
& - b \left(y_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}} \right) \\
& + R \left(x_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, X \right) \\
& - R \left(y_1, \frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, Y \right) \\
& + Q \left(\frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, X \right) \\
& - Q \left(\frac{E(t_1) H^*(x_1 - y_1) D_\xi H^*(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, Y \right) \\
& \leq 0.
\end{aligned} \tag{5.22}$$

By (5.19), we obtain $X \leq Y$. Utilizing the ellipticity of the function Q , we have

$$Q\left(\frac{E(t_1)H^*(x_1-y_1)D_\xi H^*(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{\frac{1}{2}}}, X\right) - Q\left(\frac{E(t_1)H^*(x_1-y_1)D_\xi H^*(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{\frac{1}{2}}}, Y\right) \geq 0. \quad (5.23)$$

By Property P_3 , one has

$$\begin{aligned} & b\left(x_1, \frac{E(t_1)H^*(x_1-y_1)D_\xi H^*(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}}\right) \\ & - b\left(y_1, \frac{E(t_1)H^*(x_1-y_1)D_\xi H^*(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}}\right) \\ & \geq -k_q \left| H\left(\frac{E(t_1)H^*(x_1-y_1)D_\xi H^*(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}} - w\right) \right. \\ & \quad \left. D_\xi H\left(\frac{E(t_1)H^*(x_1-y_1)D_\xi H^*(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}} - w\right) \right| H^*(x_1-y_1). \end{aligned} \quad (5.24)$$

For $t = t_1$ and $p_1 = p_2 = \frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{\frac{1}{2}}}$, it can be verified that there exist constants C_0 , C_1 and C_2 such that the following inequalities hold

$$\begin{aligned} & H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{\frac{1}{2}}} - w\right) \leq C_0, \\ & \left| D_\xi H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{\frac{1}{2}}} - w\right) \right| \leq C_1, \end{aligned}$$

and

$$\left| D_\xi^2 H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{\frac{1}{2}}} - w\right) \right| \leq C_2.$$

Therefore, we have

$$\begin{aligned} & R\left(x_1, \frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}}, X\right) - R\left(y_1, \frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}}, Y\right) \\ & = m(x_1) \operatorname{trace} \left(D_\xi H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}} - w\right) \right. \\ & \quad \otimes D_\xi H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}} - w\right) (X - Dw) \\ & \quad + H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}} - w\right) \\ & \quad \left. D_\xi^2 H\left(\frac{E(t_1)H(x_1-y_1)D_\xi H(x_1-y_1)}{(H^*(x_1-y_1)^2+\varepsilon^2)^{1/2}} - w\right) (X - Dw) \right) \end{aligned}$$

$$\begin{aligned}
& -m(y_1) \operatorname{trace} \left(D_\xi H \left(\frac{E(t_1)H(x_1 - y_1)D_\xi H(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{1/2}} - w \right) \right. \\
& \quad \otimes D_\xi H \left(\frac{E(t_1)H(x_1 - y_1)D_\xi H(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{1/2}} - w \right) (Y - Dw) \\
& \quad + H \left(\frac{E(t_1)H(x_1 - y_1)D_\xi H(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{1/2}} - w \right) \\
& \quad \left. D_\xi^2 H \left(\frac{E(t_1)H(x_1 - y_1)D_\xi H(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{1/2}} - w \right) (Y - Dw) \right) \\
& \leq C \operatorname{trace} \left(\left(\frac{m(x_1)I}{\sqrt{m(x_1)m(y_1)}I} \quad \frac{\sqrt{m(x_1)m(y_1)}I}{m(y_1)I} \right) \begin{pmatrix} X & 0 \\ 0 & -Y \end{pmatrix} \right) \\
& \leq C' H^*(x_1 - y_1)^2 E(t_1) \operatorname{trace}(U) \\
& \leq C' H^*(x_1 - y_1)^2 E(t_1) \left((H^*(x_1 - y_1)^2 + \varepsilon^2)^{-1/2} |D_\xi H^*(x_1 - y_1)|^2 \right. \\
& \quad + H^*(x_1 - y_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-1/2} |D_\xi^2 H^*(x_1 - y_1)| \\
& \quad + (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-1/2} |D_\xi H^*(x_1 - y_1)|^4 \\
& \quad + H^*(x_1 - y_1)^4 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-5/2} |D_\xi H^*(x_1 - y_1)|^4 \\
& \quad + H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-1/2} |D_\xi^2 H^*(x_1 - y_1)|^2 \\
& \quad \left. + 2H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-1/2} |D_\xi H^*(x_1 - y_1)|^2 |D_\xi^2 H^*(x_1 - y_1)| \right),
\end{aligned} \tag{5.25}$$

where $C = C(w, D_\xi H, D_\xi^2 H)$, $C' = C'(w, D_\xi H, D_\xi^2 H, k_m)$ are constants. By (5.25), one has

$$\begin{aligned}
& R \left(x_1, \frac{E(t_1)H(x_1 - y_1)D_\xi H(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, X \right) - R \left(x_2, \frac{E(t_1)H(x_1 - y_1)D_\xi H(x_1 - y_1)}{(H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}}, Y \right) \\
& \geq -C_1 H^*(x_1 - y_1)^2 E(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}} \\
& \quad - C_2 H^*(x_1 - y_1)^6 E(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{5}{2}},
\end{aligned} \tag{5.26}$$

where C_1, C_2 are constants depending on $w, D_\xi H, D_\xi^2 H, k_m$. Combining (5.22), (5.23), (5.24) and (5.26), we have

$$\begin{aligned}
& E'(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}} + \frac{\lambda}{(T - t_1)^2} - C_2 H^*(x_1 - y_1)^2 \cdot \\
& E(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}} - C_1 H^*(x_1 - y_1)^6 E(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{5}{2}} \\
& - C H^*(x_1 - y_1) \\
& \leq 0,
\end{aligned} \tag{5.27}$$

where $C = C(w, H, D_\xi H, k_q)$ is a constant. Due to

$$\begin{aligned}
& (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}(E'(t_1) - C_2 E(t_1) H^*(x_1 - y_1)^2 (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-1} - \\
& C_1 H^*(x_1 - y_1)^6 E(t_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-3} - C H^*(x_1 - y_1) (H^*(x_1 - y_1)^2 + \varepsilon^2)^{-\frac{1}{2}}) \\
& \geq (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}(E'(t_1) - C_2 E(t_1) - C_1 E(t_1) - C) \\
& \geq (H^*(x_1 - y_1)^2 + \varepsilon^2)^{\frac{1}{2}}(E'(t_1) - \beta E(t_1) - C) \\
& \geq 0,
\end{aligned} \tag{5.28}$$

where $\beta = \beta(w, D_\xi H, D_\xi^2 H, k_m)$ is a constant. By (5.27) and (5.28), we have

$$\frac{\lambda}{(T - t_1)^2} \leq 0,$$

which is a contradiction. Therefore,

$$u(x, t) - u(y, t) \leq \Phi^\varepsilon(x, y, t).$$

Taking the limit as $\varepsilon \rightarrow 0^+$, one obtains

$$u(x, t) - u(y, t) \leq E(t) H^*(x - y).$$

Switching the positions of x and y , we have

$$|u(x, t) - u(y, t)| \leq E(t) H^*(x - y), \quad t \in [0, T]. \quad \square$$

Proof of Theorem 3.14. The proof begins with a C^2 initial condition u_0 , followed by a mollifier argument to handle the Lipschitz case.

Step 1: Lipschitz estimate for the C^2 -case. For the case $u_0 \in C^2$ which is periodic and bounded, we define

$$\begin{aligned}
D_1 := \sup_{x \in \mathbb{R}^n} \{ & \zeta_q C_0 C_1 + R(x, Du_0, D^2 u_0) + Q_*(Du_0, D^2 u_0), \\
& \zeta_q C_0 C_1 - R(x, Du_0, D^2 u_0) - Q_*(Du_0, D^2 u_0) \},
\end{aligned}$$

where ζ_q, C_0, C_1 defined as Lemma 3.7 are constants. Letting $u^-(x, t) = u_0 - D_1 t$ and $u^+(x, t) = u_0 + D_1 t$, one can verify that $u_-(x, t)$ is a viscosity subsolution and $u^+(x, t)$ is a viscosity supersolution of Problem (3.2) in $\mathbb{R}^n \times [0, T]$. By Theorem 3.8, we have

$$u^-(x, t) \leq u(x, t) \leq u^+(x, t), \quad (x, t) \in \mathbb{R}^n \times [0, T],$$

which implies

$$|u(x, t) - u_0(x)| \leq D_1 t.$$

Therefore, for any $h > 0$, one obtains

$$|u(x, t + h) - u(x, t)| \leq \sup_{x \in \mathbb{R}^n} (u(x, h) - u_0) \leq D_1 h.$$

Step 2: Mollification and uniform continuity. For the general Lipschitz initial condition u_0 , we employ a standard mollifier. For each $\varepsilon > 0$, we let $u_0^\varepsilon = u_0 * \rho_\varepsilon$. Clearly, u_0^ε has the same periodicity with u_0 . Therefore, we obtain

$$|u_0^\varepsilon(x)| = |u_0 * \rho_\varepsilon(x)| = \left| \int_{\mathbb{R}^n} u_0(x-y) \rho_\varepsilon(y) dy \right| \leq \|u_0\|_{L^\infty(\mathbb{R}^n)} \int_{\mathbb{R}^n} \rho_\varepsilon(y) dy = \|u_0\|_{L^\infty(\mathbb{R}^n)},$$

which implies that u_0^ε is bounded. By [24], we have that u_0^ε is C^2 and there exists the constant $D_2 = D_2(\rho_\varepsilon, n)$, such that

$$\|Du_0^\varepsilon\|_{L^\infty(\mathbb{R}^n)} \leq E_0 \text{ and } \|D^2u_0^\varepsilon\|_{L^\infty(\mathbb{R}^n)} \leq \frac{E_0}{\varepsilon} D_2.$$

Moreover,

$$\begin{aligned} |u_0(x) - u_0^\varepsilon(x)| &= \left| u_0(x) - \int_{\mathbb{R}^n} u_0(x-y) \rho_\varepsilon(y) dy \right| \\ &= \left| \int_{\mathbb{R}^n} (u_0(x) - u_0(x-y)) \rho_\varepsilon(y) dy \right| \\ &\leq \int_{\mathbb{R}^n} |u_0(x) - u_0(x-y)| \rho_\varepsilon(y) dy \\ &\leq E_0 \int_{M(0,\varepsilon)} |y| \rho_\varepsilon(y) dy \\ &\leq \varepsilon E_0. \end{aligned}$$

Let u^ε be the viscosity solution of Problem (3.2) with initial condition u_0^ε . By Theorem 3.8, for any $\varepsilon' > 0$, we have

$$\|u^{\varepsilon'}(\cdot, t) - u^\varepsilon(\cdot, t)\|_{L^\infty(\mathbb{R}^n)} \leq \|u_0^{\varepsilon'} - u_0^\varepsilon\|_{L^\infty(\mathbb{R}^n)}.$$

Since the sequence $\{u_0^\varepsilon\}_\varepsilon$ is a Cauchy sequence and converges uniformly to u_0 , we have that $\{u^\varepsilon(\cdot, t)\}_\varepsilon$ converges uniformly to a function v . Moreover, by the uniqueness of the solution, we conclude that v is equivalent to u . By the Theorem 2.3 in [7], u is the viscosity solution of Problem (3.2) with initial condition u_0 . By Theorem 3.8, one obtains

$$\|u^\varepsilon(\cdot, t) - u(\cdot, t)\|_{L^\infty(\mathbb{R}^n)} \leq \|u_0^\varepsilon - u_0\|_{L^\infty(\mathbb{R}^n)}.$$

Consequently, for any $h > 0$, one has

$$\begin{aligned} |u(\cdot, t+h) - u(\cdot, t)| &= |u(\cdot, t+h) - u^\varepsilon(\cdot, t+h) + u^\varepsilon(\cdot, t+h) - u^\varepsilon(\cdot, t) + u^\varepsilon(\cdot, t) - u(\cdot, t)| \\ &\leq 2\|u_0^\varepsilon - u_0\|_{L^\infty(\mathbb{R}^n)} + D_1(E_0, \frac{E_0}{\varepsilon} D_2)h \\ &\leq 2\varepsilon E_0 + D_1(E_0, \frac{E_0}{\varepsilon} D_2)h. \end{aligned} \tag{5.29}$$

Due to the geometric property of equation (3.3), we have

$$D_1\left(E_0, \frac{E_0}{\varepsilon} D_2\right) \sim \frac{1}{\varepsilon}.$$

By (5.29), choosing $\varepsilon = \sqrt{\frac{h}{2E_0}}$, we obtain

$$|u(\cdot, t+h) - u(\cdot, t)| \leq \sqrt{8E_0} \cdot \sqrt{h}.$$

Thus, we have proven that u is uniformly continuous in time. $\square$

References

- [1] E. Abderrahim, D. Xavier, L. Zakaria, et al., Nonlocal Infinity Laplacian Equation on Graphs with Applications in Image Processing and Machine Learning, *Mathematics and Computers in Simulation*, vol. 102, 2014, pp. 153–163.
- [2] B. Amin, N. Mourad, L. Amine, Hybrid variable exponent model for image denoising: a nonstandard high-order PDE approach with local and nonlocal coupling, *J. Math. Anal. Appl.* 536 (2) (2024) 128245.
- [3] G. Aronsson, Minimization problems for the functional $\sup_x F(x, f(x), f'(x))$, *Ark. Mat.* 6 (1) (1965) 33–53.
- [4] G. Aubert, P. Kornprobst, *Mathematical Problems in Image Processing: Partial Differential Equations and the Calculus of Variations*, Springer Science & Business Media, 2006.
- [5] V. Balan, J. Stojanov, Finslerian-type GAF extensions of the Riemannian framework in digital image processing, *Filomat* 29 (3) (2015) 535–543.
- [6] G. Baravdish, Y. Cheng, O. Svensson, et al., Extension of p -Laplace operator for image denoising, in: *IFIP Conference on System Modeling and Optimization*, Springer International Publishing, Cham, 2015, pp. 107–116.
- [7] G. Barles, *Solutions de viscosité des équations de Hamilton Jacobi*, Mathématiques & Applications, Springer-Verlag, 1994.
- [8] G. Barles, P.E. Souganidis, Convergence of approximation schemes for fully nonlinear second order equations, *Asymptot. Anal.* 4 (3) (1991) 271–283.
- [9] M. Bertalmio, G. Sapiro, V. Caselles, et al., Image inpainting, in: *Proceedings of the 27th Annual Conference on Computer Graphics and Interactive Techniques*, 2000, pp. 417–424.
- [10] C. Bianchini, G. Ciruolo, Wulff shape characterizations in overdetermined anisotropic elliptic problems, *Commun. Partial Differ. Equ.* 43 (5) (2018) 790–820.
- [11] C. Bonamy, C. Le Guyader, Split Bregman iteration and infinity Laplacian for image decomposition, *J. Comput. Appl. Math.* 240 (2013) 99–110.
- [12] E. Carlini, S. Tozza, A scheme for the game p -Laplacian and its application to image inpainting, *Appl. Math. Comput.* 461 (2024) 128299.
- [13] V. Caselles, J.M. Morel, C. Sbert, An axiomatic approach to image interpolation, *IEEE Trans. Image Process.* 7 (3) (1998) 376–386.
- [14] L.D. Cohen, On active contour models and balloons, *CVGIP, Image Underst.* 53 (2) (1991) 211–218.
- [15] G.B. Coleman, C.H. Andrews, Image segmentation by clustering, *Proc. IEEE* 67 (5) (1979) 773–785.
- [16] M.G. Crandall, H. Ishii, L.P. Lions, User's guide to viscosity solutions of second order partial differential equations, *Bull. Am. Math. Soc.* 27 (1) (1992) 1–67.
- [17] L.S. Davis, A survey of edge detection techniques, *Comput. Graph. Image Process.* 4 (3) (1975) 248–270.
- [18] C. Elion, L.A. Vese, An image decomposition model using the total variation and the infinity Laplacian, *Comput. Imaging Vision* 6498 (2007) 261–270.
- [19] A. Elmoataz, X. Desquesnes, M. Toutain, On the game p -Laplacian on weighted graphs with applications in image processing and data clustering, *Eur. J. Appl. Math.* 28 (6) (2017) 922–948.
- [20] A. Elmoataz, M. Toutain, D. Tenbrinck, On the p -Laplacian and ∞ -Laplacian on graphs with applications in image and data processing, *SIAM J. Imaging Sci.* 8 (4) (2015) 2412–2451.
- [21] H. Ennaji, Y. Quéau, A. Elmoataz, Tug of War games and PDEs on graphs with applications in image and high dimensional data processing, *Sci. Rep.* 13 (1) (2023) 6045.
- [22] S. Esedoğlu, S.J. Osher, Decomposition of images by the anisotropic Rudin-Osher-Fatemi model, *Commun. Pure Appl. Math.* 57 (12) (2004) 1609–1626.
- [23] L.C. Evans, *Partial Differential Equations*, American Mathematical Society, 2022.
- [24] N. Forcadel, Dislocations dynamics with a mean curvature term: short time existence and uniqueness, *Differ. Integral Equ.* 21 (3–4) (2008) 285–304.
- [25] S. Fu, W. Liu, K. Zhang, et al., Semi-supervised classification by graph p -Laplacian convolutional networks, *Inf. Sci.* 560 (2021) 92–106.
- [26] G. Fu, P. Zhao, Y. Bian, p -Laplacian based graph neural networks, in: *International Conference on Machine Learning, International Conference on Machine Learning*, 2022, pp. 6878–6917.
- [27] L. Guillot, C.L. Guyader, Extrapolation of vector fields using the infinity Laplacian and with applications to image segmentation, in: *International Conference on Scale Space and Variational Methods in Computer Vision*, vol. 5567, 2009, pp. 87–99.
- [28] L. Gui, C. Li, X. Yang, Medical image segmentation based on level set and isoperimetric constraint, *Phys. Med.* 42 (2017) 162–173.
- [29] R.M. Haralick, L.G. Shapiro, Image segmentation techniques, *Comput. Vis. Graph. Image Process.* 29 (1) (1985) 100–132.
- [30] H. Ishii, On uniqueness and existence of viscosity solutions of fully nonlinear second-order elliptic PDE's, *Commun. Pure Appl. Math.* 42 (1) (1989) 15–45.
- [31] R. Jensen, Uniqueness of Lipschitz extensions: minimizing the sup norm of the gradient, *Arch. Ration. Mech. Anal.* 123 (1993) 51–74.

- [32] P. Juutinen, B. Kawohl, On the evolution governed by the infinity Laplacian, *Math. Ann.* 335 (2006) 819–851.
- [33] M. Kass, A. Witkin, D. Terzopoulos, Snakes: active contour models, *Int. J. Comput. Vis.* 1 (4) (1988) 321–331.
- [34] N. Katzourakis, *An Introduction to Viscosity Solutions for Fully Nonlinear PDE with Applications to Calculus of Variations in L^∞* , Springer, 2014.
- [35] C. Liu, Z. Qiao, Q. Zhang, Two-phase segmentation for intensity inhomogeneous images by the Allen-Cahn local binary fitting model, *SIAM J. Sci. Comput.* 44 (1) (2022) B177–B196.
- [36] C. Liu, Z. Qiao, Q. Zhang, An active contour model with local variance force term and its efficient minimization solver for multiphase image segmentation, *SIAM J. Imaging Sci.* 16 (1) (2023) 144–168.
- [37] C. Liu, Z. Qiao, Q. Zhang, Multi-phase image segmentation by the Allen-Cahn Chan-Vese model, *Comput. Math. Appl.* 141 (2023) 207–220.
- [38] B. Mebrate, A. Mohammed, Comparison principles for infinity-Laplace equations in Finsler norm, *Nonlinear Anal.* 190 (2020) 111605.
- [39] S. Minaee, Y. Boykov, F. Porikli, et al., Image segmentation using deep learning: a survey, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (7) (2021) 3523–3542.
- [40] J.M. Morel, S. Solimini, *Variational Methods in Image Segmentation: with Seven Image Processing Experiments*, Springer Science Business Media, 2012.
- [41] J.F. Ning, L. Zhang, et al., NGVF: an improved external force field for active contour model, *Pattern Recognit. Lett.* 28 (1) (2007) 58–63.
- [42] N. Otsu, A threshold selection method from gray-level histograms, *Automatica* 11 (285–296) (1975) 23–27.
- [43] Y. Peres, O. Schramm, S. Sheffield, et al., Tug-of-war and the infinity Laplacian, *J. Am. Math. Soc.* 22 (1) (2009) 167–210.
- [44] R. Ronfard, Region-based strategies for active contour models, *Int. J. Comput. Vis.* 13 (2) (1994) 229–251.
- [45] J.A. Sethian, *Level Set Methods and Fast Marching Methods: Evolving Interfaces in Computational Geometry Fluid Mechanics, Computer Vision, and Materials Science*, Cambridge University Press, 1999.
- [46] A.K. Sirohi, S. Halder, K. Kumar, et al., Enhancing robustness of graph neural networks through p -Laplacian, *arXiv Preprint*, arXiv:2511.06143, 2025.
- [47] Y. Wang, Y. Jia, L. Liu, Harmonic gradient vector flow external force for snake model, *Electron. Lett.* 44 (2) (2008) 105–107.
- [48] Y. Wang, L. Liu, H. Zhang, Z. Cao, S. Lu, Image segmentation using active contours with normally biased GVF external force, *IEEE Signal Process. Lett.* 17 (10) (2010) 875–878.
- [49] Y. Wu, Y. Wang, Y. Jia, Adaptive diffusion flow active contours for image segmentation, *Comput. Vis. Image Underst.* 117 (10) (2013) 1421–1435.
- [50] C. Xu, J.L. Prince, Snakes, shapes, and gradient vector flow, *IEEE Trans. Image Process.* 7 (3) (1998) 359–369.
- [51] C. Xu, J.L. Prince, Generalized gradient vector flow external forces for active contours, *Signal Process.* 71 (2) (1998) 131–139.
- [52] Z. Zaabouli, L. Afraites, et al., A variable $p[u]$ exponent reaction-diffusion PDE for image denoising, *J. Math. Anal. Appl.* 540 (2) (2024) 128624.
- [53] C.T. Zahn, Graph-theoretical methods for detecting and describing gestalt clusters, *IEEE Trans. Comput.* 100 (1) (1971) 68–86.
- [54] Z. Zhang, C. Duan, T. Lin, et al., GVFOM: a novel external force for active contour based image segmentation, *Inf. Sci.* 506 (2020) 1–18.